

The Neuron-specific IIS/FOXO Transcriptome in Aged Animals Reveals Regulatory Mechanisms of Cognitive Aging

Reviewed Preprint

Revised by authors after peer review.

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Reviewed preprint version 2

May 14, 2024 (this version)

Reviewed preprint version 1

March 27, 2024

Sent for peer review

January 17, 2024

Posted to preprint server

December 23, 2023

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Abstract

Cognitive decline is a significant health concern in our aging society. Here, we used the model organism *C. elegans* to investigate the impact of the IIS/FOXO pathway on age-related cognitive decline. The *daf-2* Insulin/IGF-1 receptor mutant exhibits a significant extension of learning and memory span with age compared to wild-type worms, an effect that is dependent on the DAF-16 transcription factor. To identify possible mechanisms by which aging *daf-2* mutants maintain learning and memory with age while wild-type worms lose neuronal function, we carried out neuron-specific transcriptomic analysis in aged animals. We observed downregulation of neuronal genes and upregulation of transcriptional regulation genes in aging wild-type neurons. By contrast, IIS/FOXO pathway mutants exhibit distinct neuronal transcriptomic alterations in response to cognitive aging, including upregulation of stress response genes and downregulation of specific insulin signaling genes. We tested the roles of significantly transcriptionally-changed genes in regulating cognitive functions, identifying novel regulators of learning and memory. In addition to other mechanistic insights, comparison of the aged vs young *daf-2* neuronal transcriptome revealed that a new set of potentially neuroprotective genes is upregulated; instead of simply mimicking a young state, *daf-2* may enhance neuronal resilience to accumulation of harm and take a more active approach to combat aging. These findings suggest a potential mechanism for regulating cognitive function with age and offer insights into novel therapeutic targets for age-related cognitive decline.

eLife assessment

This **fundamental** study investigates the transcriptional changes in neurons that underlie loss of learning and memory with age in *C. elegans*, and how cognition is maintained in insulin/IGF-1-like signaling mutants. The presented evidence is **compelling**, utilizing a cutting-edge method to isolate neurons from worms for genomics that is clearly conveyed with a rigorous experimental approach. Overall, this study supports that older *daf-2* worms maintain cognitive function via mechanisms that are unique from younger wild type worms, which will be of great interest to neuroscientists and researchers studying ageing.

<https://doi.org/10.7554/eLife.95621.2.sa3>

Introduction

The loss of cognitive function is a rising problem in our aging society. A 2008 study estimated that at least 22.2% (about 5.4 million) individuals over the age of 71 in the United States have at least mild cognitive impairment^{1,2,3}. Furthermore, global dementia cases are predicted to triple from an estimated 57.4 million cases in 2019 to 152.8 million cases in 2050^{4,5}. As most industrialized countries are experiencing a rapid increase in the proportion of the aged population, understanding and potentially preventing the underlying issues of neuronal structural and behavioral decline associated with aging is crucial for societal health.

C. elegans is an excellent model system for studying neuronal aging, given its tractable genetics, short lifespan, and simple nervous system⁶. Most importantly, *C. elegans* experiences rapid loss of learning and memory with age⁷; by Day 4 of adulthood, all long-term associative memory ability is lost, and by Day 8, *C. elegans* cannot carry out associative learning or short-term associative memory⁷ – despite the fact that these worms can still move and chemotaxis perfectly well. That is, with age worms first lose long-term memory ability (by Day 4), then short-term memory and learning ability (Day 6-8), then chemotaxis (Day 10-12), then motility (Day 16)^{7,8}. Because learning and memory decline extremely early, we consider worms that are only a week old to already be “cognitively aged,” despite the fact that they can chemotax and move well, and will continue to live for another one to two weeks. Therefore, we can examine neurons from these 7-8 day-old adults to explore the causes of these cognitive declines in animals that are otherwise quite healthy. Many human neuronal aging phenotypes and genes of interest for mammalian neuronal function are conserved in *C. elegans*⁹, making discoveries in *C. elegans* possibly applicable to humans.

The Insulin/IGF-1-like signaling (IIS)/FOXO pathway was first discovered to play a role in longevity in *C. elegans*. The lifespan of *daf-2*/Insulin/IGF-1 receptor mutants is twice that of wild-type animals¹⁰, and this lifespan extension requires the downstream Forkhead box O (FOXO) transcription factor DAF-16¹⁰. DAF-16/FOXO controls the expression of many genes that contribute to longevity, including stress response, proteostasis, autophagy, antimicrobial, and metabolic genes¹¹. As a conserved regulator, the IIS/FOXO pathway also regulates longevity in *Drosophila*, mice, and humans^{12–15}. In addition to regulating lifespan, the IIS pathway regulates neuronal function via the FOXO transcription factor. In particular, *C. elegans* IIS/*daf-2* mutants display DAF-16-dependent improved learning, short-term memory, and long-term memory⁷. While both young and old *daf-2* adult worms display increased learning and memory relative to wild-type, the duration of this extension is not known, and the mechanisms by which *daf-2* mutants maintain neuronal function in older worms is not yet understood. Compared to

wild-type worms, *daf-2* mutants better maintain maximum velocity⁸, motility^{16,17}, neuromuscular junctions, the ability to regenerate axons^{18,19}, and neuronal morphology with age^{20–22}. In particular, we previously showed that while *daf-2* has lower observed motility on food²³, this apparent is due to its high levels of a food receptor, ODR-10⁸, and its downregulation reveals the much higher mobility of *daf-2* animals⁸, even on food, in addition to its much higher and maintained maximum velocity with age. Previously, we found that *daf-2* worms also extend learning beyond wild-type's ability⁷, but the full duration of this extension with age was not known. That is, exactly how late in life *daf-2* mutants can still learn and remember, and whether this is proportional to their lifespan extension, was not previously determined.

We previously performed neuron-specific RNA-sequencing in young (Day 1) adult *C. elegans* and identified neuron-specific targets²⁴; genes upregulated in *daf-2* mutant neurons are distinct from those in the whole animal, and we found that these neuronal genes are necessary for the observed improvements in memory and axon regeneration in *daf-2* mutant worms. However, whether *daf-2* uses the same or different genes in young and old worms to improve and maintain cognitive function with age is unknown. Recent datasets using whole-animal single-cell RNA-seq have been generated for wild-type and *daf-2* worms, and these are sufficient for whole-body aging and pseudobulk analyses^{25–27}, but we have found that those data are not deep enough to use specifically for in-depth analysis of neurons, which can be difficult to gather from whole animals. Other data are from larval stages and cannot be extrapolated to aging adults²⁸. To identify the transcriptional differences in the aging nervous system that might contribute to the loss of neuronal function with age in wild-type worms and the differences responsible for the extended abilities of *daf-2* animals, here we performed RNA sequencing on FACS-isolated neurons of aged (Day 8) wild-type and IIS/FOXO mutants. To further investigate the role of the neuronal IIS/FOXO pathway, we identified genes both upregulated by the IIS/FOXO pathway, and genes that are differentially expressed in *daf-2* mutants with age. We found that *daf-2* differentially-regulated genes in the aged neurons are different from young neurons; in fact, many of these Day 8 *daf-2* vs *daf-16*; *daf-2* upregulated genes are stress response and proteolysis genes that may promote neuronal function and health. We then used functional assays to assess the contributions of *daf-2*-regulated genes to learning and memory. Our results suggest that *daf-2*'s neuronal targets in older worms are required to maintain neuronal functions with age, suggesting that additional and alternative mechanisms are at work in these aged mutants from their young counterparts.

Results

Wild-type neurons lose their neuronal function and identity with age

Previously, we found that cognitive abilities in *C. elegans*, including learning, short-term memory, and long-term memory, all decline with age⁷. Moreover, neuronal morphology and regeneration ability are also impaired with age^{18,20–22}. However, how these phenotypes are regulated at the molecular level in aging neurons remains to be systematically characterized. Therefore, we were interested in first identifying gene expression changes with age in wild-type neurons to characterize the normal physiological aging process. Before choosing timepoints to assess neuronal transcriptome changes, we carried out associative learning and short-term associative memory assays⁷ as we have previously described^{7,29–31}. Briefly, well-fed worms are starved for one hour, then re-fed while exposed to the neutral odorant butanone for one hour; a choice assay between butanone and control immediately after training tests associative learning, while a choice assay after one hour of recovery on food-only plates tests short-term associative memory⁷. Adult Day 1 worms are fully developed, young, and healthy, while wild-type Day 7–8 worms, although still in their mid-life, have completely lost their learning and short-term memory abilities already by Day 7⁷ (**Fig. 1a**), thus we consider them “aged” for the purposes of

understanding loss of cognitive ability. Therefore, we reasoned that a comparison of adult wild-type Day 1 neurons with wild-type neurons that are at least aged Day 7-8 should reveal changes with age that result in loss of cognitive function.



Figure 1.

Identifying neuronal aging targets in WT worms using neuron-specific RNA-sequencing.

(a) Wild-type learning and 1hr memory results on Day 1 and Day 7. Learning and memory results are represented as learning index (LI). Details of the LI calculation are explained in the methods. Learning, N = 10, memory, N = 5. ****: $p < 0.01$. Student's t-test. (b) PCA plot for Day 1 (orange) and Day 8 (blue) neuronal bulk RNA-seq samples. (c) Volcano plot comparing age-associated differentially-expressed genes in WT neurons. Genes downregulated with age (orange) and upregulated with age (blue) were obtained by neuron-specific RNA sequencing of adult wild-type animals with neuron-specific GFP expression. Adjusted p-value < 0.001 , $\log_2(\text{Fold-change}) > 2$. N = 6 biological replicates per age. 1146 genes were significantly downregulated with age (higher in young neurons) and 2016 genes were upregulated with age (higher in old neurons) (d) Tissue prediction scores for genes higher in young neurons. (e) GO terms of genes that decline with age in wild-type neurons. Synaptic and signaling GO terms are enriched in neuronal genes. p-value calculated using hypergeometric distribution probability. (f) Comparison of whole-body higher-in-young genes and neuronal higher-in-young genes. GO Terms and representative genes were performed using g:Profiler software. P-value of overlapping regions were calculated using a hypergeometric calculator. (g) Normalized reads of *ins-6*, *unc-4*, *mec-7*, *fol-2*, *fbf-1*, and *madd-4*, in Day 1 and Day 8 neurons in our dataset. P-adjusted values were calculated from DESeq2 software. Box plots: center line, median; box range, 25-75th percentiles; whiskers denote minimum-maximum values.

To identify genes that regulate age-related morphological and functional decline in wild-type neurons, we performed neuron-specific transcriptomic analysis using our previous FACS neuronal isolation method²⁴ on six biological replicates each of Day 1 and Day 8 adult wild-type worms, where 100,000 GFP+ cells were collected for each sample (Fig. S1a–c). Because we previously found that whole-worm analyses mask changes found specifically in neurons²⁴, to complement our aging neuron studies, we also carried out RNA-sequencing analyses of aging whole worms (Fig. S2a–e), which we found is dominated by changes in the extracellular matrix (Fig. S2b), stress response/pathogen genes (Fig. S2c) and the alimentary system (intestine) (Fig. S2d), overshadowing neuronal changes.

Principal components analysis of the FACS-isolated neuron RNA-seq samples indicated that they are well separated by age (Fig. 1b), and downsampling analysis³² (Fig. S1e) suggested that we have sequenced to saturation, with an average of 41,636,463 uniquely-counted reads and detected the expression of 19725 coding and non-coding genes ($\log_{10}(\text{TPM}) > 0.5$) (Fig. S1d).

Enrichment analysis of genes that are differentially expressed with age (Fig. 1c–e) suggested that neuronal sorting and sequencing were successful because the sequenced genes are enriched for neuronal genes such as *mec-7*, *mec-12*, and *twk-49*, as expected, and less enriched for all other major tissues. Tissue enrichment analysis of differentially-expressed genes suggested that aging neurons lose genes most expressed in the nervous system and neurons (Fig. 1d), as one might expect. Gene ontology (GO) analysis suggested that genes declining with age in neurons encode proteins important in neuronal function (Fig. 1e, f), including synaptic proteins (e.g., *srh-59*, *rab-3*, *sng-1*, *sup-1*), potassium channels (e.g., *egl-23*, *twk-7*, *twk-49*, *ncs-5*), and transmembrane transporters (e.g., *fol-2*, *ccb-2*, *unc-79*, *exp-1*). The decrease in expression of these genes during aging may indicate that neurons are losing their identity and their ability to perform neuronal functions, such as signal transduction and axonal transport, and correlates with the behavioral and morphological declines observed in aging wild-type worms.

Comparing whole-worm sequencing and neuron-specific sequencing (Fig. 1f), we found that genes involved in metabolic processes decline with age only in the body, and genes encoding structural proteins, lipid localization, and muscle system processes decline with age in both the

body and in neurons, while neurons specifically lose genes that are associated with neuronal function, including synaptic proteins, neuropeptide signaling, and other neuron functions, correlating with neuronal loss of function with age. Together, these results indicate that neurons harbor many unique age-related changes that could be overshadowed in the whole-worm transcriptome but are revealed by neuron-specific sequencing.

Many genes that are more highly expressed in young neurons are known to be specific to a subset of neurons. *ins-6*, an insulin-like peptide specific to the ASI, ASJ, and AWA neurons²⁸ that regulates longevity³³ and aversive learning³⁴, is significantly downregulated with age (**Fig. 1g**). *srd-23*, a serpentine receptor located at the AWB neuron cilia³⁵, also decreases expression with age (**Fig. S1f**). Furthermore, various genes specific to sensory neurons (*txt-12*, *flp-33*), touch neurons (*mec-7*), and motor neurons (*unc-4*) decline in expression with age (**Fig. 1g**, **Fig. S1f**).

Previous studies showed that loss of genes including *ins-6*³⁴, *mec-7*³⁶, *unc-4*, *fol-2*¹⁹, *madd-4*³⁷, and *fbf-1*³¹ lead to behavioral dysfunction in motility and chemosensory abilities; therefore, the decreased expression of these neuron type-specific genes with age may impact the function of individual neurons and disrupt neural circuit communication, ultimately contributing to the declines in behavior observed during aging (**Fig. 1g**).

As neurons age, genes that increase in expression, while assigned to the nervous system (**Fig. 2a**) are not specific for neuron function; instead, aged wild-type neurons express higher levels of many predicted F-box genes with predicted proteasome E3 activity (e.g., F-box proteins *fbxa-158*, *fbxb-51*, *pes-2.1*, and SKp1-related proteins *skr-12*, *skr-6*). Some transcription regulation (e.g., *ced-13*, *tbx-43*, *nhr-221*, *end-1*), and chromatin structure and function (e.g., *his-54*, *dot-1.2*, *jmjd-3.2*, *hil-7*, *utx-1*) genes also increase with age (**Fig. 2b**), even though neurons appear to lose their neuron-specific transcriptional identity with age.

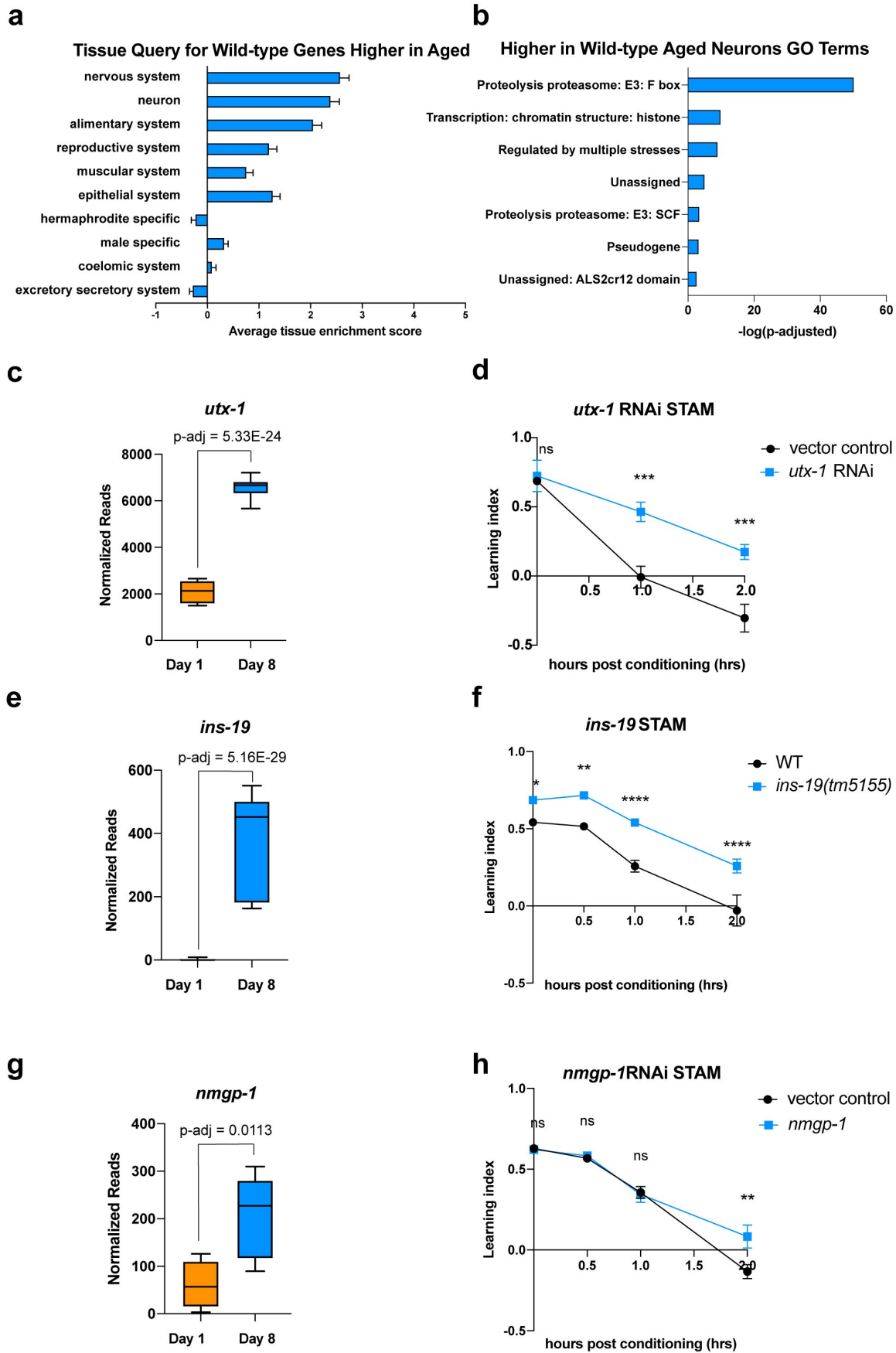


Figure 2.

Genes that increase with age cause behavioral defects.

(a) Tissue prediction score for wild-type genes expressed at higher levels in aged worms. (b) GO terms of genes expressed higher in aged neurons highlight transcription regulation and proteolysis. GO term analysis was done using Wormcat 2.0. (c) Normalized reads of *utx-1* on Day 1 and Day 8. (d) Short-term associative memory (STAM) assay shows that adult-only *utx-1* knockdown improves 1hr and 2hr memory of wild-type worms on Day 2. (e) Normalized reads of *ins-19* on Day 1 and Day 8. (f) *ins-19* mutation improves learning and memory in STAM on Day 3 of adulthood. (g) Normalized reads of *nmgp-1* on Day 1 and Day 8. (h) *nmgp-1* RNAi knockdown improves memory in STAM on Day 2. P-adj value of normalized count change generated from DESeq2 analysis. (c,e,g) Box plots: center line, median; box range, 25-75th percentiles; whiskers denote minimum-maximum values. Normalized reads and adjusted p value calculated using the DESeq2 software. Each dot represents one sequencing replicate. (d,f,h) N = 5 plates in each behavioral experiment. Representative result of 2 biological repeats is shown. *: p<0.05. **: p<0.01. ***: p<0.001. ****: p<0.0001. Two-way ANOVA with Tukey's post-hoc analysis.

One ongoing discussion about changes during aging is how to interpret an increase in expression with age. There are two main models for genes that increase their expression with age and have a resulting impact on function: that they rise with age to compensate for lost function (“compensatory”) and therefore promote function, or that their expression is deleterious to function and only rises with age through dysregulation. If a gene is compensatory, then its knockdown would abrogate learning and memory, even in young animals. If a gene’s function is harmful to neurons, reducing its expression might be beneficial to the worm, even in young animals. (Of course, there may be other scenarios in which a gene with multiple functions may be detrimental for some behaviors but beneficial for other physiological roles.) To test this hypothesis, we reduced expression of a small set of highly-upregulated candidate genes in categories that might function in a compensatory manner.

These include *utx-1*, a histone demethylase known to play a role in development³⁸ and lifespan in worms^{39–41}, and whose homolog has been implicated in cognition in mammals^{42,43}; *ins-19*, an insulin-like peptide; and *nmgp-1*, a neuronal glycoprotein involved in chemosensation⁴⁴. In each case, we see that gene expression is significantly higher in old than in young neurons (**Fig. 2c, e, g**). If a gene increases expression to benefit neurons, we would expect to see no difference in memory in young animals where there is no defect; by contrast, if the increase of a gene is deleterious, we would expect to see an improvement in behavior when knocked down, even in young animals. We performed adult-only neuron-specific RNAi knockdown to prevent any possible deleterious effects caused by changes during development, which largely takes place in early larval stages; testing in young adult animals is logical because there is no memory in aged wild-type, so any deleterious effect of knocking down a potentially compensatory gene in an aged would not result in a change. For all behavioral assays, we first prioritized significantly-changed genes with high fold-change, and then those with mammalian homologs.

We found that 48 hrs (L4-Day 2) of adult-only knockdown of *utx-1* increases 1hr and 2hr memory (**Fig. 2d**), the loss-of-function mutation of *ins-19* increases both learning and memory (**Fig. 2f**) and the adult-only knock-down of *nmgp-1* extends memory at 2 hours (**Fig. 2h**). That is, in each of these cases, reduction of these genes did not impair memory, as loss of a compensatory function would appear; rather, loss of these age-upregulated genes improved wild-type memory. These results indicate that at least some neuronal genes that increase with age can have a negative impact on learning and memory, as demonstrated by the improvement of memory when knocked down, even in young animals. While it is still possible that some upregulated genes may act in a compensatory manner, the simplest model is that at least some are actively deleterious for learning and memory. We previously observed that for genes that play a role in complex

behaviors like learning and memory, the loss of single genes can have a large impact on these complex behaviors⁶⁰, unlike the additive roles of longevity-promoting genes¹¹. Therefore, one mechanism by which wild-type worms lose their learning and memory functions with age is not just by loss of neuronal gene expression, as one might expect, but also by dysregulation of expression of genes that can negatively impact learning and memory.

***daf-2* mutants maintain learning and memory with age**

We previously found that *daf-2* animals have extended motility (Maximum Velocity) that correlates with and predicts their extension of lifespan⁸. Additionally, not only do young *daf-2* worms have better memory than wild-type worms, but *daf-2* mutants also maintain learning and memory better with age^{7,24}. However, the duration of this improvement was unknown. To determine the proportion of life that worms can learn and remember, we tested wild-type, *daf-2*, and *daf-16;daf-2* worms for their learning and associative memory ability every day until these functions were lost. We found that while wild-type worms lose their learning and short-term memory abilities by Day 7-8 (**Fig. 1a**, **Fig. 3a, b**), learning and memory span were significantly extended in *daf-2* mutants (**Fig. 3a, b**); thus, a comparison of *daf-2* neurons with wild-type neurons at Day 8 should reveal differences relevant to cognitive aging. The extension of learning and memory is dependent on the FOXO transcription factor DAF-16 (**Fig. 3a**); in fact, while *daf-16;daf-2* mutants still have the ability to learn for a few days, these mutants are completely unable to carry out any memory ability, even on Day 1. Thus, learning ability, which is similar in wild-type and *daf-2;daf-16* mutants, is mechanistically distinct from short-term memory ability³¹. *daf-2* worms maintained learning ability until Day 19 and short-term (1hr) memory ability until Day 15, more than twice the duration of wild-type worms, while *daf-16;daf-2* worms exhibit no short-term memory ability, even on Day 1 of adulthood (**Fig. 3b**). Our data suggest that the learning span-to-lifespan (**Fig. S3c**) and memory span-to-lifespan ratios in *daf-2* worms were similar to or slightly higher than that of wild-type worms (**Fig. 3c, d**), indicating that *daf-2* mutants maintain cognitive function for at least proportionally as long as wild-type worms do. Thus, *daf-2* mutants maintain their higher cognitive quality of life longer than wild-type worms, while *daf-16;daf-2* mutants spend their whole lives without memory ability (**Fig. 3d**), in contrast to claims that *daf-2* mutants are less healthy than wild-type or *daf-16* worms²³. Additionally it should be noted that because our choice assays distinguish motility function from learning and memory function²⁹, the improvements in memory with age shown by *daf-2* mutants relative to wild type are distinct from *daf-2*'s improvements in motility that we previously showed⁸. Therefore, we are interested in these genes that might contribute to the extended cognitive function that *daf-2* worms demonstrate.

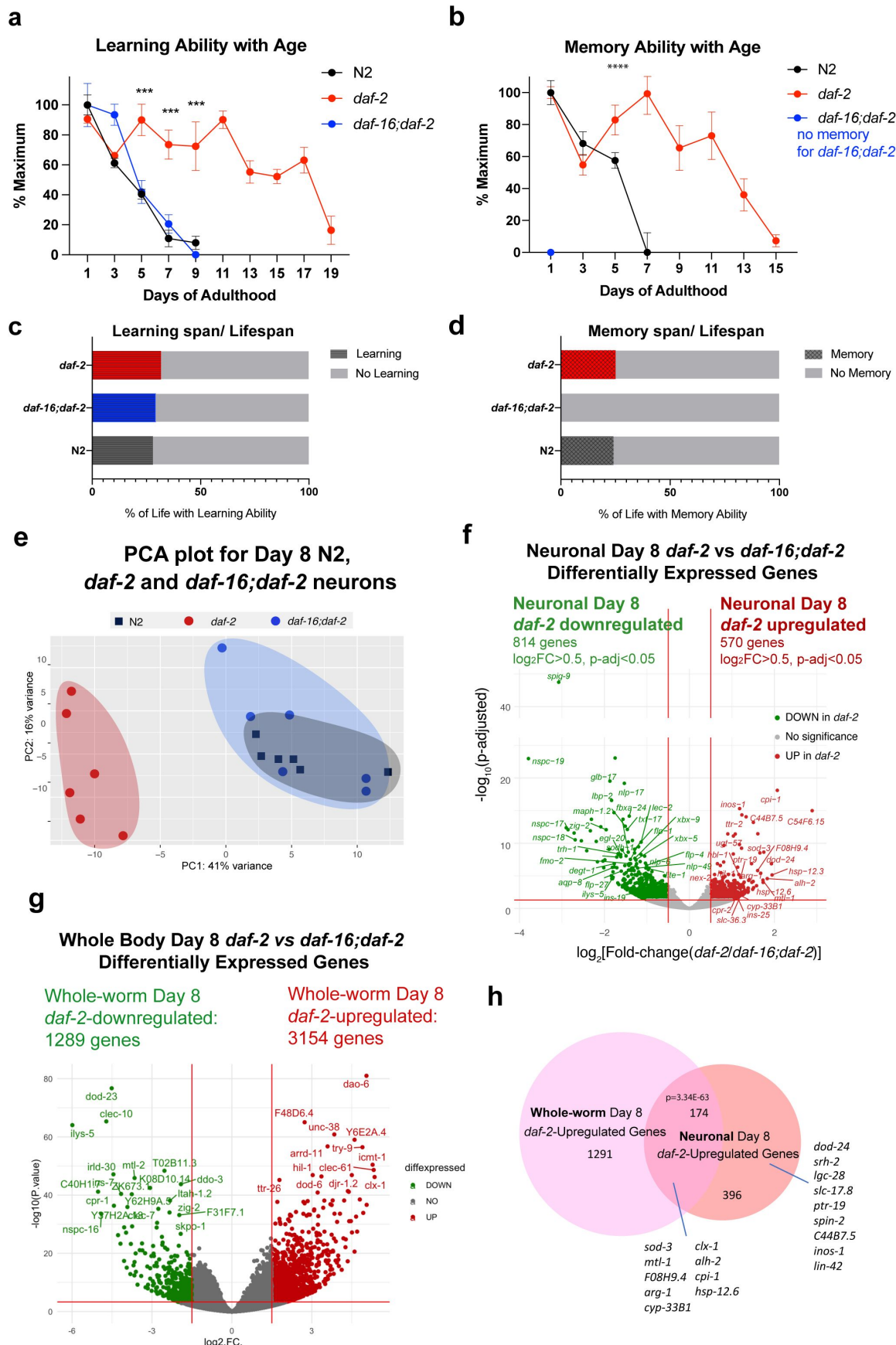


Figure 3.

Identifying neuronal IIS/FOXO targets in aged worms using neuron-specific RNA-sequencing.

(a) *daf-2* mutants show better learning maintenance with age compared to N2 and *daf-16;daf-2* worms. N = 10 plates in each condition. (b) *daf-2* mutants show better memory maintenance with age compared to N2 worms. *daf-16;daf-2* worms do not have 1hr memory on Day 1 of adulthood. N = 10 plates in each condition. (c-d) *daf-2* mutants have a slightly larger learning span/lifespan ratio and memory span/lifespan ratio than N2 (wild type). Lifespan shown in Figure S3c. (e) PCA plot of Day 8 N2, *daf-2*, and *daf-16;daf-2* neuronal RNA sequencing results. (f) Volcano plot of neuronal *daf-2*-regulated, *daf-16*-dependent up- and downregulated genes on adult Day 8 (Adjusted p-value < 0.05, $\log_2(\text{Fold-change}) > 0.5$, N = 6 biological replicates per strain). 570 genes were significantly upregulated and 814 genes were downregulated in *daf-2* neurons compared with *daf-16;daf-2*. (g) Volcano plot of whole-worm *daf-2* vs *daf-16;daf-2* differentially-expressed genes during aging. 3154 genes are higher in *daf-2*, 1289 genes are higher in *daf-16;daf-2* ($\log_2[\text{Fold-change}(\text{daf-2 vs daf-16;daf-2})] > 1.5$, p-adjusted < 0.01). (h) Comparison of neuronal and whole-worm Day 8 *daf-2* differentially-expressed genes (overlap p = 3.34E-63, hypergeometric test). Neuron-specific and shared *daf-2* upregulated genes with the highest fold-changes are labeled.

Aging IIS/FOXO neurons express stress-resistance genes to maintain neuronal function with age

To identify genes that may improve memory and slow cognitive aging in long-lived *daf-2* mutants, we compared the transcriptional profiles of Day 8 FACS-isolated neurons from *daf-2* animals with Day 8 FACS-isolated wild-type and *daf-16;daf-2* neurons; by Day 8, wild-type and *daf-16;daf-2* worms have already lost their learning and memory ability, but *daf-2* worms still maintain their cognitive functions (Fig. 3a, b). It should be noted that wild-type worms still have normal chemotaxis and motility at Day 8⁷, and there is a separation of several days between the loss of cognitive functions and the loss of motility⁷; therefore, comparison of the neuronal transcriptomes of *daf-2* with wild type and *daf-16;daf-2* at this age should specifically highlight genes that are required for learning and memory rather than other functions.

The PCA of the *daf-2*, *daf-16;daf-2*, and wild-type neuronal Day 8 transcriptomes (Fig. 3e) indicates that aged *daf-16;daf-2* mutant neurons are similar to aged wild-type neurons, correlating well with their similarly worsened cognitive functions at this age; that is, at a transcriptomic level, aged (Day 8) wild-type neurons and aged *daf-16;daf-2* neurons are similar, which is echoed by their shared inability to carry out learning and memory by Day 8 of adulthood (Fig. 3ea, b). By contrast, the transcriptomes of aged *daf-2* mutant neurons are distinct from both aged wild-type and aged *daf-16;daf-2* neuron transcriptomes, just as the cognitive abilities of *daf-2* are much greater than wild type or *daf-16;daf-2* at this age.

Downsampling analysis shows that our sequencing depth is sufficient to saturate the detectable differential expression (Fig. S3g, h). We obtained an average of 47,233,119 counted reads per sample (Supplemental Table 5) and detected the expression of 16,488 coding and non-coding genes (Fig. S3e).

We identified 570 upregulated and 814 downregulated genes in Day 8 *daf-2* neurons compared to Day 8 *daf-16;daf-2* neurons (Fig. 3f). A large fraction of the downregulated genes in Day 8 *daf-2* vs *daf-16;daf-2* neurons are “nematode-specific peptide family” (*nspc*-) genes of unknown function (Fig. 3f). While the *daf-2* vs *daf-16;daf-2* changes in whole worms largely replicated the results from our previous studies of young animals^{11,45} (Fig. 3g, Fig S6g), comparison of the *daf-2* vs *daf-16;daf-2* differential transcriptional changes in Day 8 whole worms and Day 8 neurons

reveal shared (174 genes) and neuron-specific gene expression changes (396 genes; *dod-24*, *srh-2*, *lin-42*, etc.) (Fig. 3h, Fig. S4c). Not surprisingly, previously-identified genes from whole-worm *daf-2* vs *daf-16*; *daf-2* and N2 (e.g., *sod-3*, *mtl-1*, *cpi-1*, *hsp-12.6*, etc.) that play roles in both neurons and other tissues even in Day 1 *daf-2* mutants appear in the shared list. (Some neuron-specific Day 8 *daf-2*-upregulated genes have not been reported to be expressed in neurons previously (e.g., *spin-2*), further suggesting the value of transcriptomic analyses of isolated neurons in mutant backgrounds at this age.)

Many genes upregulated in Day 8 *daf-2* neurons relative to *daf-16*; *daf-2* are related to stress responses, including heat stress (e.g., *hsp-12.6*, *hsp-12.3*, *F08H9.4/hsp*), oxidative stress (e.g., *sod-3*), and metal stress genes (e.g., *mtl-1*); and proteolysis (e.g., *cpi-1*, *cpr-2*, and *tep-1*). The upregulation of these genes may perform neuroprotective functions, as their homologs in mammals have been shown to do (Table 1). Specifically, 36 of the top 100 upregulated genes have identified orthologs or identified domains with known function, of which 32 of (89%) have functions in promoting neuronal health. These mammalian homologs protect neurons against protein aggregation and harmful metabolites (e.g., *cpi-1*, *alh-2*, *ttr-41*, *gpx-5*)^{46–50}, maintain synaptic organization and neuronal homeostasis (e.g., *dod-24*, *ptr-19*, *plep-1*)^{51–53}, facilitate neuronal injury repair (e.g., *F08H9.4*, *sod-3*)^{54,55}, and maintain normal neuronal function (e.g., *lgc-28*, *slc-36.3*, *lin-42*)^{56–59}. Together, these genes may help maintain *daf-2*'s neuronal health and protect neurons from accumulation of environmental harm during aging.

We found that about a third of the *daf-2*-upregulated genes were shared between the *daf-2* vs *daf-16*; *daf-2* analysis and the *daf-2* vs N2 analysis (338 genes) (Fig. S5). Of the unshared genes, the *daf-2*-maintained genes that are specific to the *daf-2* vs N2 comparison are bZIP transcription factors, including *zip-5*, *zip-4*, *atf-2*, and proteasome components (Figure S6D).

These results indicate that other transcription factors may participate in regulating *daf-2* functions in aged neurons in addition to the *daf-16*/FOXO transcription factor.

IIS/FOXO transcriptomic changes are necessary for *daf-2* mutant's improved neuronal functions

We were interested not only in the genes that remained upregulated with age, but also in genes that might have increased with age in the high-performing *daf-2* mutants. That is, are there genes that increase in expression in *daf-2* mutants that are necessary or beneficial for their continued high performance with age? Some of the Day 8 *daf-2* vs wild-type or *daf-16*; *daf-2* upregulated genes are also Class 1 DAF-16-dependent genes¹¹ (*sod-3*, *hsp-12.3*, *fat-5*, and *mtl-1*, *hil-1*, and *dao-2*). However, many more genes were differentially expressed in Day 8 *daf-2* vs *daf-16*; *daf-2* neurons from our Day 1 data²⁴ (Fig. 4a, Fig. S4d). Of the “new” genes – that is, genes upregulated specifically in neurons of Day 8 vs Day 1 of *daf-2* vs *daf-16*; *daf-2* – many have mammalian homologs that have been shown to play neuroprotective roles, by protecting against aggregation proteins and harmful metabolites, maintaining synaptic organization, neuronal homeostasis, or neuronal activity, or facilitating neuronal injury repair (see Table 1 for specific references).

If the upregulated genes in aged *daf-2* neurons are responsible for the extended memory span of *daf-2* mutants, knocking down those genes should block older *daf-2* mutants' memory functions. Therefore, we tested the effect of RNAi knockdown of the top fold-change candidate genes on *daf-2*'s memory in aged adults. We chose Day 6 for testing because by then, like on Day 8, wild-type worms have already lost their learning and most memory abilities, but *daf-2* worms retain normal cognitive functions, and this time point avoids the increased naïve chemotaxis that we observe in older *daf-2* animals. As shown in Fig. 4b–c, we selected these significantly differentially-expressed candidate genes based on their ranking in fold-change.

Gene Name	Full Name	log2(FC)	P-adj	Mammalian Ortholog	Ortholog Full Name	Inferred Function
Neuroprotective against Neurodegenerative Diseases						
<i>cpi-1</i>	Cysteine Protease Inhibitor 1	2.07	7.80E-19	CST3	Cystatin C	Protease inhibitor, suppresses AD pathology ¹
<i>alh-2</i>	Aldehyde deHydrogenase 2	1.83	2.65E-05	ALDH1A1	Aldehyde dehydrogenase 1	Expressed in dopaminergic neurons. Regulates dopamine release in Parkinson's Disease ²
<i>ttr-41,45,2</i>	TransThyretin-Related family domain 41,45,2	1.68	3.98E-06			Inhibits Aβ fibril formation, and suppresses the AD pathology ³
<i>cyp-33B1</i>	Cytochrome P450 family 33B1	1.34	2.04E-03	CYP2J2	Cytochrome P450 2J2	Protective against Parkinson's Disease through altered metabolism ^{4,5}
<i>spin-2</i>	SPINster (Dm lysosomal permease) homolog 2	1.27	6.20E-04	SPNS2	Spinster homolog 2	Sphingosine-1-phosphate Transporter, neuroprotective in AD ⁶
<i>gpx-5</i>	Glutathione PeroXidase 5	1.27	3.99E-04	GPX3,5,6	glutathione peroxidase 3,5,6	Protects again lipid peroxidation, protects against neurodegeneration ^{7,8}
<i>cpr-2</i>	Cysteine PRotease related 2	1.25	5.01E-03	CTSB	Cathepsin B	Lysosomal Protease, Involved in Aβ and APP protein degradation ⁹
<i>djr-1.2</i>	DJ-1 (mammalian transcript'l regulator) Related 1.2	1.09	3.90E-03	PARK7	Parkinsonism associated deglycase	Neuroprotective against Parkinson's Disease; Prevents accumulation of harmful metabolites ¹⁰
Synaptic Organization Maintenance						
<i>dod-24</i>	Downstream Of DAF-16 (regulated by DAF-16) 24	1.93	1.39E-07		Cub-like Domain Containing Protein	Clustering of neurotransmitter receptor proteins ¹¹
<i>ptr-19,15</i>	PaTched Related family 19,15	1.21	1.72E-05	PTCHD1,3,4	Patched domain-containing 1,3,4	Synaptic organization, autism risk factor ^{12,13}
<i>hbl-1</i>	HunchBack Like (fly gap gene related) 1	1.16	6.47E-06	hb	Hunchback (fly)	Regulate synapse number and locomotor circuit function ¹⁴
<i>cutl-4</i>	CUTiclin-Like 4	1.08	2.74E-02	pio	Piopia (fly)	ECM protein for axonal growth and synapse formation ¹⁵
<i>Iron-2</i>	eLRR (extracellular Leucine-Rich Repeat) ONLY 2	1.06	8.70E-05	LGI1,2	Leucine-Rich Glioma Inactivated protein 1	Modulation of trans-synaptic proteins. Protection against seizure ¹⁶
Neuronal Homeostasis Maintenance						
<i>mocs-1</i>	MOLybdenum Cofactor Sulfurase 1	1.05	1.17E-04	MOCOS	Molybdenum cofactor sulfurase	Regulation of redox homeostasis and synaptogenesis. Down in ASD ¹⁷
<i>plep-1</i>	Plugged Excretory Pore 1	1.12	2.92E-03	MFSD11	Major facilitator superfamily domain 11	Putative SLC solute carrier protein, involved in brain energy homeostasis ¹⁸
<i>cky-1</i>	CKY homolog 1	1.08	1.58E-04	NPAS4	Neuronal PAS Domain Protein 4	Calcium-dependent transcription factor, neuronal homeostasis maintenance ^{19,20}
Neuronal Injury Repair facilitation						
<i>F08H9.4, hsp-12.3,12.6</i>	small HSP domain-containing protein	1.94	7.33E-06	HSPB2	Heat-shock Protein Beta 2	Facilitates PNS injury regeneration, suppresses inflammation ²¹
<i>sod-3</i>	SOD superoxide dismutase 3	1.66	3.05E-09	SOD2	superoxide dismutase2	Converts superoxide to the less reactive hydrogen peroxide (H ₂ O ₂). Protects neurons from injury. ²²
Normal Neuronal Activity Maintenance						
<i>lgc-28</i>	Ligand-Gated ion Channel 28	1.38	7.29E-04	CHRNA6,3	Neuronal acetylcholine receptor subunit alpha-6,3	Nicotinic receptor. Regulates cognitive functions and addiction ^{23,24} 3/15/2024 12:10:00 AM
<i>F22B7.9</i>		1.33	8.91E-15	METTL23	methyltransferase like 23	Interacts with GABPA; disruption causes intellectual disability ^{25,26}
<i>fat-5</i>	FATty acid desaturase 5	1.31	3.40E-03	SCD5	StearoylCoA Desaturase-5	Neuronal Cell Proliferation and Differentiation ²⁷
<i>slc-36.3</i>	SLC (SoLute Carrier) homolog 36.3	1.25	2.88E-03	SLC36A4	Solute Carrier Family36 Member4	amino acid transporter, transports Trp, involved in kynurenic acid pathway ²⁸
<i>lin-42</i>	abnormal cell LINEage 42	1.15	1.66E-04	PER1,2	Period 1,2	Phosphorylates CREB, modulates CREB-mediated memory consolidation ²⁹
<i>ctsa-1.1</i>	CaThepSin A homolog 1.1	1.07	4.97E-05	CTSA	Lysosomal Ser carboxy-peptidase Cathepsin A	Involved in normal neuronal development ^{30,31}
<i>gsnl-1</i>	GeISolIN-Like 1	1.06	2.93E-04	AVIL	advillin	Facilitates somatosensory neuron axon regeneration ³²

Table 1.

List of top *daf-2* vs *daf-16*; *daf-2* upregulated genes with orthologs that have neuroprotective functions

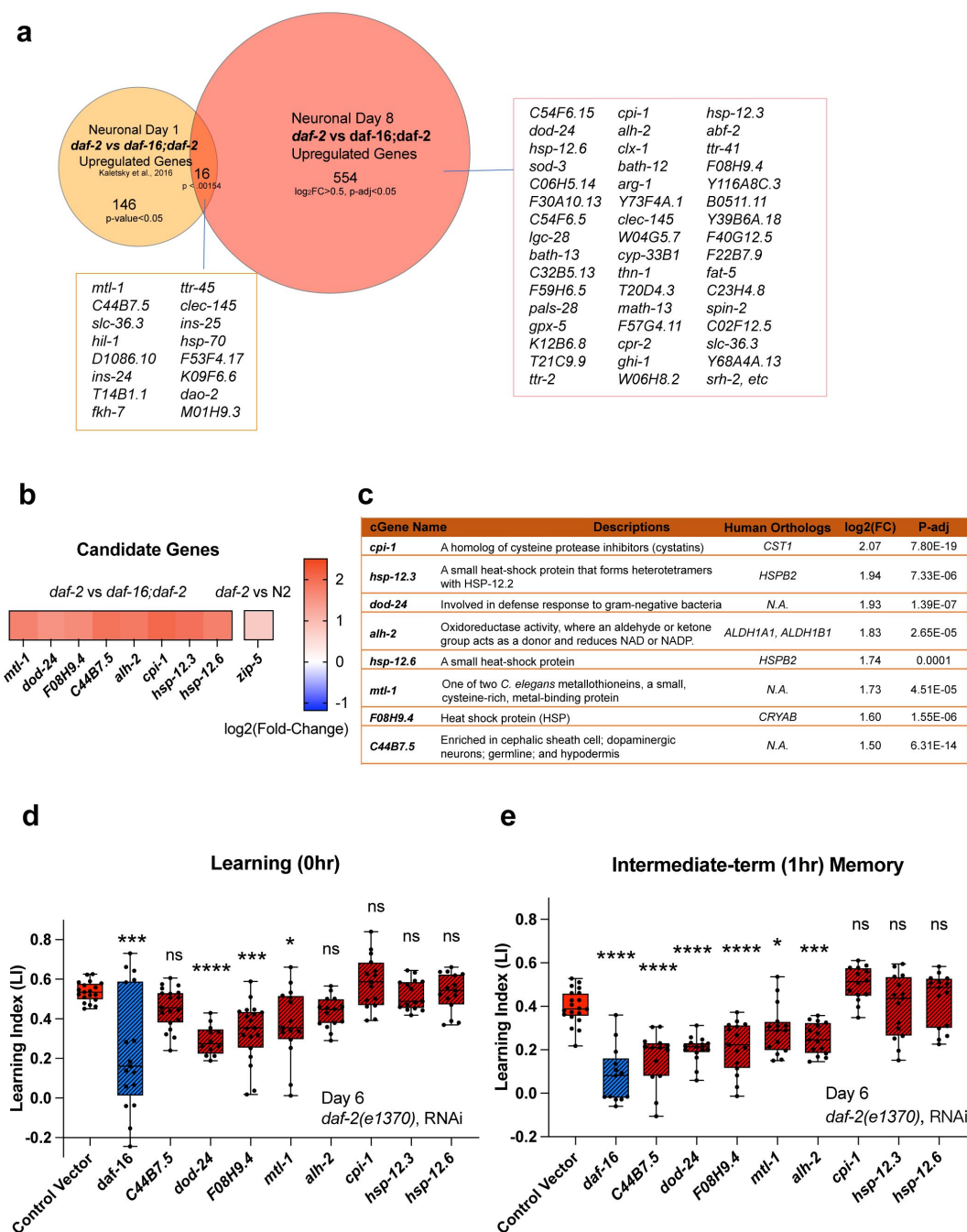


Figure 4.

Neuronal IIS/FOXO aging targets regulate memory decline with age in *daf-2* worms.

(a) Comparison of neuronal Day 1 and Day 8 *daf-2* vs *daf-16*; *daf-2* upregulated genes. All shared genes and top Day 8-specific *daf-2* upregulated genes are labeled. (b) *daf-2*-regulated fold-change profile of candidate genes. All candidates are upregulated in *daf-2* mutants. (c) Description of candidate genes. log₂(Fold-change) and p-adjusted values from the *daf-2* vs *daf-16*; *daf-2* comparison unless stated otherwise. (d) Candidate gene knockdown effects on Day 6 adult *daf-2* learning (0hr after conditioning). Two candidate genes, *dod-24* and *F08H9.4*, show a significant decrease in learning ability. N = 5 plates in each condition, merged results of 3 biological repeats shown. (e) Candidate gene knockdown effects on Day 6 adult *daf-2* short-term memory (1hr after conditioning). *C44B7.5*, *dod-24*, *F08H9.4*, *mtl-1*, and *alh-2* showed significant decreases in memory. N = 5 plates in each condition, the representative image of 3 biological repeats shown. (d-e) *: p < 0.05. **: p < 0.01. ***: p < 0.001. ****: p < 0.0001. One-way ANOVA with Dunnett's post-hoc analysis. Box plots: center line, median; box range, 25-75th percentiles; whiskers denote minimum-maximum values.

Previously, we have found that the top significantly differentially-expressed genes (by fold-change) are most likely to have strong effects on function, while less differentially-changed genes have less of an effect^{11,24,60}, therefore we prioritized genes that are significantly different and the most highly expressed in *daf-2* mutants compared to *daf-16;daf-2* mutants for subsequent testing (Fig. 4c). *daf-2* worms, including neurons, are susceptible to RNA interference^{24,61}. Of the eight candidate genes we tested, the reduction of three of them (*F08H9.4*, *mtl-1*, and *dod-24*, originally classified as a Class II gene with proposed immune activity) significantly decreased *daf-2*'s learning ability on Day 6 (Fig. 4d). Those genes plus reduction of two additional genes (*C44B7.5* and *alh-2*) affected 1hr memory (Fig. 4e) in Day 6 *daf-2* mutants. That is, knockdown of the heat shock-related gene *F08H9.4*, the innate immunity gene *dod-24*, aldehyde dehydrogenase *alh-2* and previously uncharacterized gene *C44B7.5* are required to some degree for *daf-2*'s extended memory ability. The reduction of the metal stress gene *mtl-1*, which is expressed in neurons as well as the rest of the body, had a slight effect on learning and memory.

One caveat of these experiments is that, while we found these genes through the isolation of neurons from aged worms and subsequent RNA-sequencing, the knockdown of the genes and its effects are not necessarily neuron-autonomous; however, *alh-2* and *F08H9.4* were reported to only be expressed in neurons and the cephalic sheath cell⁶², and *C44B7.5* and *dod-24*, while expressed more broadly, were not upregulated in *daf-2* in the whole-worm analysis (Fig. 3f), therefore their effects are most likely neuron-autonomous. In fact, *dod-24* is one of the original Class 2 *daf-2*-downregulated genes from whole-animal analyses, suggesting that *dod-24*'s increase in expression is specifically in neurons, therefore the effect of its knockdown is most likely to be neuron-autonomous.

Together, these data suggest that the specific genes that are differentially regulated in Day 8 *daf-2* mutants may aid in slowing neuronal function decline and behavioral changes associated with aging. Furthermore, memory maintenance with age might require additional genes that function in promoting stress resistance and neuronal resilience, which were not previously uncovered in analyses of young animals.

Discussion

Although it has been shown previously that *daf-2* worms maintain various functions with age, how long they can maintain learning and memory with age, and the genes that might be responsible for these extended neuronal functions, have not been previously explored. Here we have found that *daf-2* worms maintain learning and memory abilities proportional with (or even slightly beyond) their degree of lifespan extension, underscoring *daf-2*'s improved healthspan⁸.

To understand how memory is lost with age and retained in insulin/IGF-1-like signaling mutants, we have characterized the neuronal transcriptomes of aged wild-type worms and IIS (*daf-2*) and IIS/FOXO (*daf-16;daf-2*) mutants (Fig. 5). We found that wild-type neuronal aging is characterized by a downregulation of neuronal function genes and an upregulation of proteolysis genes and transcriptional and epigenetic regulators, which together may help explain the loss of neuronal identity and function with age. We also identified the transcriptomic profile accompanying *daf-2*'s extended learning and memory span. Specifically, *daf-2* neurons maintain higher expression of stress response genes and predicted neuronal homeostasis functions (Table 1), which may help make them more resistant to environmental adversities and age-related decline. We also identified genes responsible for wild-type worms' worsened learning and memory with age.

By employing a FACS-based neuron-sorting technique, we selectively analyzed adult neuron-function-related genes and investigated their aging process, which is not easily discernible through whole-worm sequencing^{25,27} (Fig. S6). Sequencing many biological repeats of

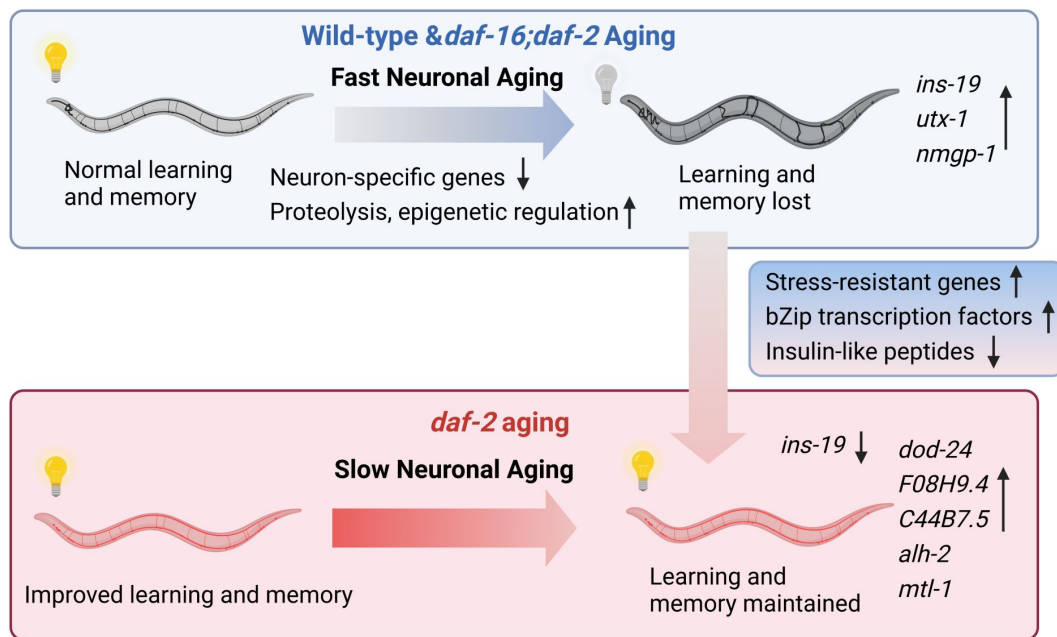


Figure 5.

Aged *daf-2* neurons upregulate neuroprotective genes to maintain improved cognitive behaviors.

During normal neuronal aging, neuron-specific genes decrease in expression, while proteolysis and epigenetic regulators are upregulated, resulting in neuron dysfunction and cognitive function loss. In aged *daf-2* mutants, upregulation of neuroprotective genes including *dod-24*, *F08H9.4*, *C44B7.5*, *alh-2*, and *mtl-1* contribute to *daf-2*'s improved cognitive function. The diagram was generated using Biorender.

aging neurons to high depth with ribosomal RNA depletion allowed us to detect a larger number of genes compared to other neuron-related bulk and single-cell sequencing profiles²⁶, providing a deep transcriptomic dataset of aged wild-type, IIS mutant, and IIS/FOXO mutant neurons. Our analysis allowed us to identify differentially-expressed genes that are known to be expressed in at a small number of neurons, even for low-abundance genes. Notably, our sequencing results uncovered genes previously not known to be expressed in neurons that remained undetected in other datasets. Moreover, we revealed the involvement of known neuronal genes in the aging process, such as *ins-6* and *srd-23*. We hope that this dataset will become a valuable resource for detecting new candidates in neuronal aging.

For example, *dod-24*, which we observed to be upregulated in *daf-2* neurons and required for *daf-2*'s extended memory, was downregulated in the *daf-2* whole-worm transcriptome (**Fig. S4c**). *dod-24* has been traditionally classified as a Class II gene that is downregulated in *daf-2* worms and upregulated by *daf-16* RNAi treatment^{11,45}. Functionally, it has been shown to be an innate immunity gene upregulated during pathogen infection^{63–65}, and its whole-body reduction has been shown to extend the lifespan of wild-type animals¹¹. However, here we find that *dod-24* is beneficial in the nervous system and required for *daf-2*'s extended learning and memory in aged worms. This intriguing contrast between the whole-worm transcriptome and the neuron-specific transcriptome suggests that some genes may have distinct regulatory roles in the nervous system, necessitating a more precise approach beyond whole-worm transcriptomics.

Using this neuron-specific sequencing profile of aged cells, we identified key pathways that change during neuron aging. Our sequencing of aged neurons uncovered active transcriptomic alterations during aging, resulting in not just transcriptional silencing but also upregulation of various pathways. We found that the Day 8 *daf-2* vs *daf-16*;*daf-2* neuronal differentially-expressed genes that we have newly discovered here differ from the neuronal Day 1 *daf-2* vs *daf-16*;*daf-2* dataset we previously obtained²⁴. These Day 8 differentially-expressed genes are not canonical neuronal genes, such as receptors and ion channels; instead, there are more metabolic and proteolytic genes whose protein orthologs have been shown to be neuroprotective. All top 50 genes and 90% of the top 100 genes with identified mammalian orthologs have been shown to be essential to neuronal functions in mammals (**Table 1**). These results indicate that instead of simply mimicking a young state, *daf-2* may enhance neuron's resilience to accumulation of harm and take a more active approach to combat aging. These changes suggest that *daf-2*'s extended memory maintenance may require different mechanisms than function in young animals; *daf-2* may maintain neuronal function not just by retaining a youthful transcriptome, but also by increasing the expression of genes that promote resilience, such as stress-response genes and proteolysis inhibitors.

In addition to examining aging in wild-type and IIS/FOXO mutants independently, our results further linked the normal aging process to altered gene regulation in the IIS pathway. *utx-1*, *nmgp-1*, and *ins-19* increase in expression in aged neurons, and we found that their reduction improved memory, indicating that at least some of the genes whose expression rises with age can have a negative impact on normal cognitive functions, rather than acting in a compensatory manner. *utx-1* is an H3K27me3 histone demethylase we found to be higher in wild-type aged neurons, but it is also involved in the IIS pathway. The downregulation of *utx-1* has been shown to regulate development³⁸ and promote longevity^{39–41}, and its mammalian homolog has been implicated in regulating cognitive abilities^{42,43}. The longevity response of *utx-1* depends on *daf-16*^{39–41}. The loss of *utx-1* decreases methylation on the *daf-2* gene, thus increasing DAF-16's nuclear localization, mimicking a *daf-2* mutation³⁹. This example of the crosstalk between normal aging and IIS/FOXO mutants offers valuable insights into modifying the aging process for enhanced longevity and cognitive health.

We found that the insulin-like peptide *ins-19* was upregulated in aged neurons and was downregulated in aged *daf-2* neurons, and its downregulation in wild-type worms extended memory span. Insulin-like peptides play crucial roles as receptor ligands (in both agonist and antagonist roles) for DAF-2, and we have found them to be downregulated in *daf-2* mutants compared with *daf-16;daf-2* mutants, possibly creating a feedback loop that dampens the insulin signaling pathway, as was previously shown for *ins-7* and *ins-18*^{11,66}. These peptides exhibit diverse functions in development, dauer formation, and longevity^{11,66,70}. Notably, certain insulin-like peptides have been linked to neuronal activities, such as the regulation of aversive learning by the two antagonistic peptides *ins-6* and *ins-7*³⁴, and reduced long-term learning and memory by *ins-22* RNAi⁶⁰. In our study, the expression changes of *ins-19* during wild-type aging and in *daf-2* mutants provide an example of how longevity mutants can reverse wild-type transcriptional changes during aging, ultimately reducing behavioral and functional decline.

Summary of mechanistic insights

Our analysis of transcriptomes from isolated aged Day 8 neurons of wild-type, *daf-2*, and *daf-16;daf-16* mutants reveal several major mechanistic insights. Specifically, we found that wild-type neurons lose their neuronal identity through a combination of the loss of neuron-specific function genes with age, and the concomitant dysregulated increase in non-neuronal genes with age. Furthermore, at least a fraction of the top-upregulated genes with age can play deleterious roles; that is, they rise with age, and their knockdown improves function, even in young animals. This argues against the idea that all of these genes are playing a compensatory role with age.

We also found that the knockdown of individual top-ranked genes that function in learning and memory can have a large impact - like removing a cog of a machine. This is in contrast to our earlier findings regarding gene reduction in lifespan, where most cellular longevity processes regulated by DAF-16 activity appear to be additive, and therefore loss of individual major genes downstream of DAF-2 and DAF-16 have at most a 5-10% impact¹¹. Several of these genes we found to be required for *daf-2*'s age-related improvement in learning and memory - namely *dod-24*, *F08H9.4*, *C44B7.5*, and *alh-2* - were previously not associated with memory function. Finally, these genes are distinct from the set of upregulated Day 1 *daf-2* vs *daf-16;daf-2* genes; how they each individually maintain neuronal function better with age will be interesting to dissect.

Conclusions

Beyond our sequencing analysis, we have established links between genomics, function, and behavior. We also identified several new genes required for *daf-2*'s age-related improvement in learning and memory, shedding light on their neuron-specific roles. These additional findings further suggest that neuronal sequencing datasets can be used to identify functional candidates and pathways during the aging process. By bridging the gap between transcriptomic landscapes, genetic regulation, and functional outcomes, our study provides a greater understanding of the mechanisms underlying neuronal aging, providing insights into development of aging interventions.

Methods

Strains and worm cultivation

N2 (wild type), OH441: *otIs45(unc-119::GFP)*, CQ295: *otIs45(unc-119::GFP);daf-2(e1370)*, CQ296: *otIs45(unc-119::GFP);daf-16(mu86);daf-2(e1370)*, LC108: *uIs69(myo-2p::mCherry + unc-119p::sid-1)*, CQ705: *daf-2(e1370) III*, 3X outcrossed, CQ745: *daf-2(e1370) III*; *vIs69[pCFJ90(Pmyo-2::mCherry + Punc-119::sid-1)] V*, QL188: *ins-19(tm5155) II*, CX3695: *kyIs140(str-2::GFP + lin-15(+))*, CQ461: (*daf-*

2(*e1370*);*Pmec-4::mCherry*) and CQ501: (*daf-2 (e1370)*;*zip-5(gk646)*;*Pmec-4::mCherry*). Strains were grown on high-growth media (HGM) plates seeded with *E. coli* OP50 bacteria using standard methods ⁷¹[71](#).

Tissue-specific isolation

For neuronal isolation, five plates of fully-grown worms from HG plates were synchronized by hypochlorite treatment, eggs spread on seeded plates to hatch, and at least five plates/replicate were grown to L4 on HGM plates until transferred to HGM plates with FUDR to avoid progeny contamination. This gives us ~6000 healthy Day 8 worms to sort. Neuron isolation and Fluorescent-activated cell sorting were carried out as previously described ²⁴[24](#),⁶²[62](#).

Briefly, worms were treated with 1000uL lysis buffer (200 mM DTT, 0.25% SDS, 20 mM HEPES pH 8.0, 3% sucrose) for 6.5 mins to break the cuticle. Then worms were washed and resuspended in 500uL 20mg/mL pronase from *Streptomyces griseus* (Sigma-Aldrich). Worms were incubated at room temperature with mechanical disruption by pipetting until no whole-worm bodies were seen, and then ice-cold osmolarity adjusted L-15 buffer(Gibco) with 2% Fetal Bovine Serum (Gibco) were added to stop the reaction. Prior to sorting, cell suspensions were filtered using a 5um filter and sorted using a FACSVantage SE w/ DiVa (BD Biosciences; 488nm excitation, 530/30nm bandpass filter for GFP detection). Sorting gates were determined by comparing with age-matched, genotype-matched non-fluorescent cell suspension samples. Fluorescent neuron cells were directly sorted into Trizol LS. 100,000 GFP+ cells were collected for each sample.

RNA extraction, library generation and sequencing

We used standard trizol-chloroform-isopropanol method to extract RNA, then performed RNA cleanup using RNeasy MinElute Cleanup Kit (Qiagen). RNA quality was assessed using the Agilent Bioanalyzer RNA Pico chip, and bioanalyzer RIN>6.0 samples were observed before library generation. 2ng of RNA was used for library generation using Ovation SoLo RNA-Seq library preparation kit with AnyDeplete Probe Mix-C. *elegans* (Tecan Genomics) according to the manufacturer's instructions ⁷²[72](#). Library quality and concentration was assessed using Agilent Bioanalyzer DNA 12000 chip. Samples were multiplexed and sequencing were performed using NovaSeq S1 100nt Flowcell v1.5 (Illumina).

Data processing

FastQC was performed on each sample for quality control analysis. RNA STAR package was used for mapping paired-end reads to the *C. elegans* genome ce11 (UCSC Feb 2013) using the gene model ws245genes.gtf. Length of the genomic sequence around annotated junctions is chosen as read length -1. 50-70% reads were uniquely mapped. Reads uniquely mapped to the genome were then counted using htseq-count (mode = union). DESeq2 analysis was then used for read normalization and differential expression analysis on counted reads ⁷³[73](#). Genes with a $\log_{10}$ TPM >0.5 were considered as detected and genes with a $\log_2$ (fold-change) > 0.5 and p-adjusted <0.05 are considered differentially expressed in further analysis. Gene ontology analysis were performed using gprofiler⁷⁴[74](#) or WormCat 2.0⁷⁵[75](#) and category 2 was selected to show. Tissue query was performed on top 500 highest fold-change genes, using worm tissue query website (www.worm.princeton.edu)⁶²[62](#), and only major systems were selected in the analysis.

Learning and Memory Experiments

We performed Short-Term Associative Memory (STAM) experiments as previously described ⁷[7](#),²⁹[29](#). Briefly, we used five plates of synchronized adult worms/sample to perform each experiment. One plate of worms was used to test naïve chemotaxis assay without conditioning, while the other 3 plates were washed into M9, and washed 3 additional times to get rid of the bacteria. These worms are starved for 1hr to prime them for food uptake. Then these worms are transferred to conditioning plates with NGM plates seeded with OP50 and 10% butanone stripes on

the lid for 1hr to perform conditioning. After conditioning, worms are either transferred from the conditioning plate directly to the chemotaxis plates to assess learning, or transferred to holding plate for 1hr or 2hrs to assess for memory. After staying on holding plates for 1hr or 2hrs, worms are then transferred onto chemotaxis plates to assess for short-term memory. Chemotaxis assays were performed by transferring worms onto chemotaxis plates with 1uL 10% butanone and 1uL ethanol spots separated by 8cm on a 10cm NGM plate. Worms who have reached either butanone spot or the ethanol spot are paralyzed by the 1uL 7.5% NaN₃ on these spots. For each timepoint, 5 chemotaxis plates are used to minimize the variation of the outcome. We performed this chemotaxis assay to butanone on naïve and appetitive-trained worms at different time points to assess change in preference to butanone. Chemotaxis index is calculated as (# of worms at butanone-# of worms at ethanol)/(total # of worms - # of worms at origin).

Learning index is calculated by subtracting trained chemotaxis index with naïve chemotaxis index.

For learning and memory span assays, we obtained synchronized worms from hypochlorite-treated eggs. Synchronized worms were washed onto 5'-fluorodeoxyuridine (FUdR) at L4 and maintained on FUdR plates by transferring to new plates every 2 days. 1 Day Prior to experiments, worms are washed onto fresh HG plates without FUdR to avoid change in behavior caused by FUdR. To verify that FUdR has no effect on short-term memory, we compared worms with and without FUdR (**Fig. S6d** [↗](#)), and found no differences. For *utx-1* and *nmgp-1* RNAi experiments, synchronized L4 wild-type worms were washed onto HGM plates with carbenicillin and IPTG and seeded with HT115 RNAi bacteria containing the RNAi constructs from the Ahringer Library. For *daf-2* upregulated candidates' RNAi experiments, synchronized L4 *daf-2* RNAi-sensitive worms were washed onto HGM plates added with carbenicillin, FUdR, and isopropyl-β-D-thiogalactopyranoside (IPTG) and seeded with HT115 bacteria containing RNAi constructs generated from the Ahringer RNAi Library, then were transferred onto fresh RNAi plates every 2 days until Day 6. 1 Day Prior to experiments, worms are transferred onto plates without FUdR.

Quantitative and Statistical Analysis

All experimental analysis was performed using Prism 8 software. Two-way ANOVA with Tukey post-hoc tests were used to compare the learning curve between control and experimental groups. One-way ANOVA followed by Dunnett post-hoc tests for multiple comparisons was performed to compare learning or 2hr memory between various treatment groups and control. Chi-square test was performed to compare the neuron morphology change between young and aged AWC neurons. All GO term analysis were performed using Wormcat 2.0 software with Bonferroni corrected adjusted p-values. Venn diagram overlaps were compared using the hypergeometric test. Differential expression analysis of RNA-seq were performed using DESeq2 algorithm and adjusted p-values were generated with Wald test using Benjamini and Hochberg method (BH-adjusted p values). Additional statistical details of experiments, including sample size (with n representing the number of chemotaxis assays performed for behavior, RNA collections for RNA-seq, and the number of worms for microscopy), can be found in the methods and figure legends. Regression analyses were performed using sklearn packages. Correlations were calculated using the SciPy packages.

Data availability

Sequencing reads are deposited at NCBI BioProject under accession number PRJNA999305.

Code availability

Analysis codes generated from this study are available upon request.

Resource availability

Further information and requests for resources and reagents should be directed to and will be fulfilled by Coleen T. Murphy (ctmurphy@princeton.edu).

Acknowledgements

We thank the Caenorhabditis Genetics Center (CGC) for strains, WormBase (version WS289) for information, Jasmine Ashraf, William Keyes, Yichen Weng, and Titas Sengupta for help with the experiments, R. Arey and other Murphy lab members for an early replicate of *daf-2* learning and memory with age, members of the Murphy Lab for input on the manuscript, Christina DeCoste, Katherine Rittenbach and the Flow Cytometry Facility for cell sorting assistance, Wei Wang and the Genomics Facility for sequencing assistance, Lance Parsons, Bruce Wang, and Chen Dan for insights on data analysis, and *Biorender.com* [for](#) schematic design. C.T.M. is the Director of the Simons Collaboration on Plasticity in the Aging Brain (SCPAB), which supported the work, and the Glenn Center for Aging Research at Princeton. Y.W. and S.Z. are supported by China Scholarship Council (CSC). K.M. is supported by HHMI Gilliam Fellows Program.

Author Contributions

C.T.M., R.K., S.Z., and Y.W. conceived and designed experiments. Y.W. and S.Z. performed RNA-sequencing experiments. Y.W., S.Z., and K.M. performed behavioral assays. Y.W. performed imaging experiments. Y.W., S.Z., K.M., S.L., R.K., and C.T.M. analyzed and interpreted results. Y.W. and C.T.M. wrote the paper.

Competing interests

The authors declare no conflict of interest.

Supplemental Figures

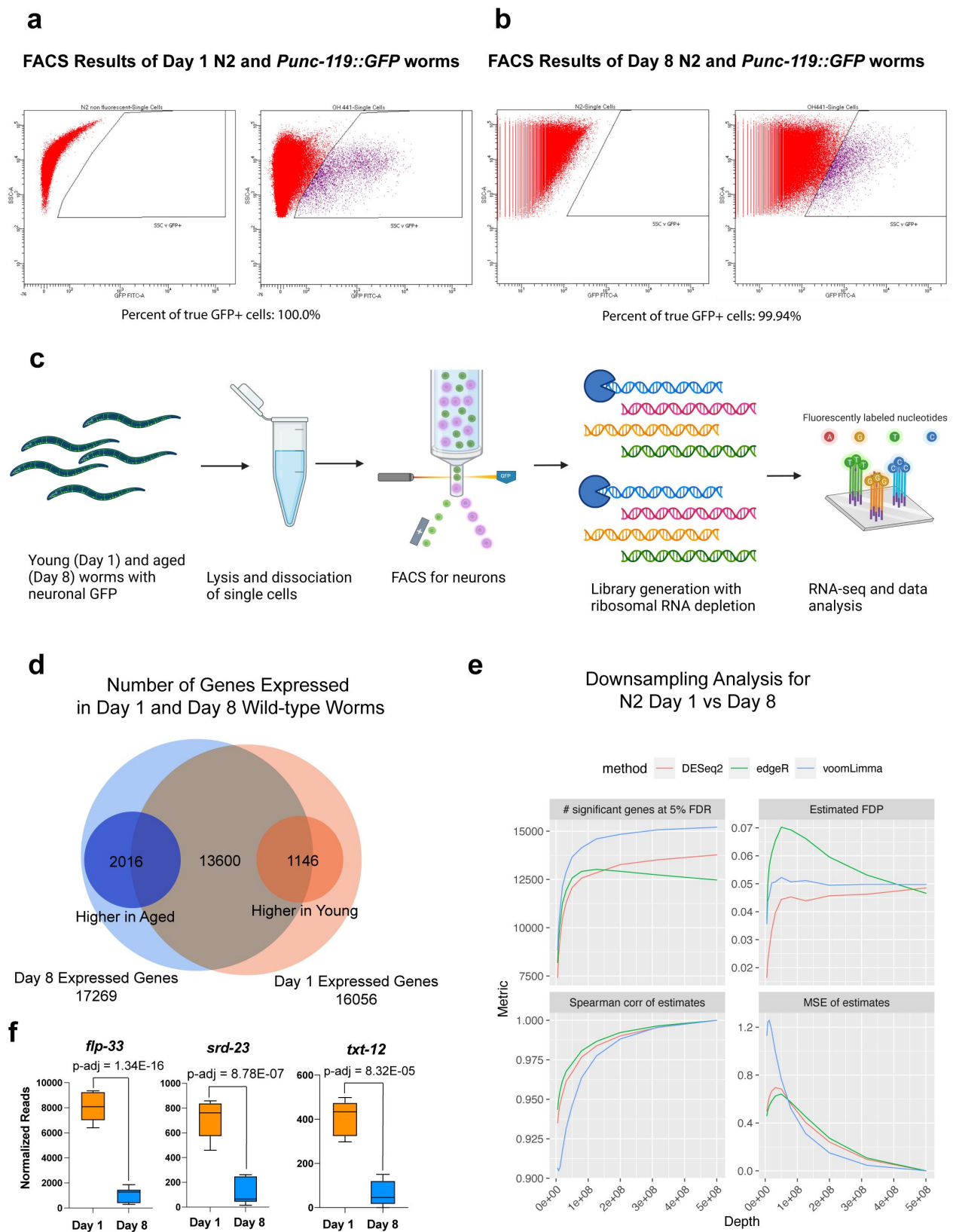


Figure S1.

Comparison with recent sequencing datasets.

(a) Comparison with CeNGEN (L4 stage) gene expression data²⁸ shows high correlation, with many genes only detected in our isolated neuron bulk-sequencing dataset (orange; wild-type Day 1 neuron sequencing data used in the comparison). Genes with an average $\log_2(\text{TPM}) > 0.5$ are considered detected. (b) Venn diagram showing genes detected in this dataset (orange), CeNGEN dataset (brown) and both (gray); only wild-type Day 1 neuron sequencing data were used in the comparison. (c-d) Comparison with gene expression data from the age-matched neuron cells from the Roux et al., dataset²⁷ shows high correlation, with many genes only detected in our bulk-sequencing dataset. (e-f) Venn diagram showing genes detected in this dataset (orange or blue), Roux et al., dataset (brown) and both (gray). Only age-matched neuron cells from the Roux et al., dataset was used for gene expression analysis. (g) Correlation between our whole-worm Day 8 *daf-2* and *daf-16;daf-2* sequencing and Class I and Class II gene rank from Tepper et al., 2013⁴⁵. High correlation indicates consistency between our isolated neuron RNA-sequencing results with former microarray results, despite the differences in approaches and ages. (h) STAM with and without FUDR shows that 3 Days on FUDR does not affect learning and memory ability.

Table S1: Whole-worm DEseq2 results

Table S2: Neuronal WT Day 1 vs Day 8 DEseq2

Table S3: Neuronal Day 8 *daf-2* vs *daf-16;daf-2* DEseq2

Table S4: Neuronal Day 8 *daf-2* vs N2 DEseq2

Table S5: Neuronal Day 8 *daf-16;daf-2* vs N2 DEseq2

Table S6: Number of sequencing reads

Table S7: Raw behavioral data

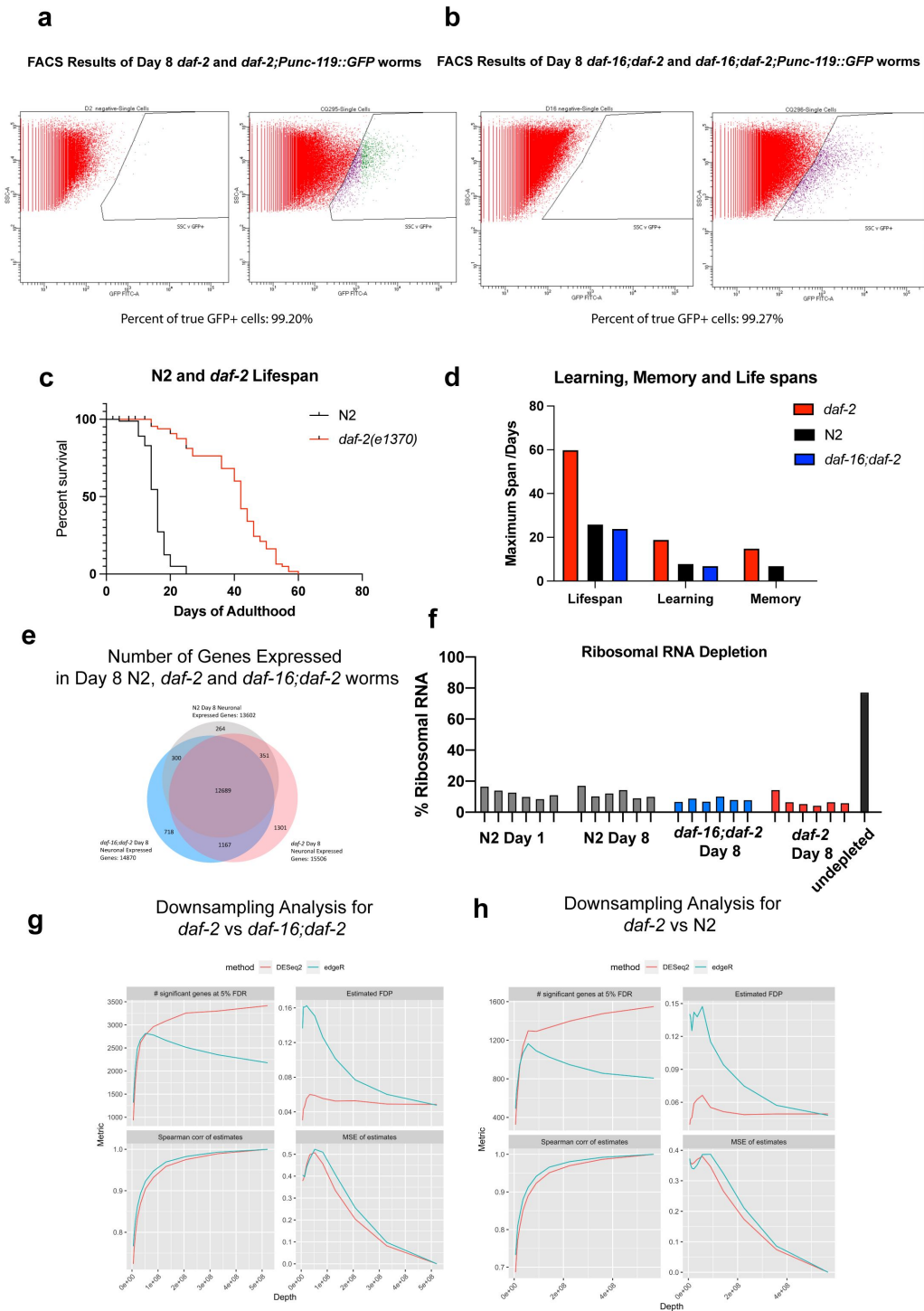


Figure S3.

Whole-worm RNA-sequencing identifies whole-body changes during aging.

(a) Volcano plot of Day 1 vs Day 8 differentially-expressed genes during aging. 264 genes are expressed at higher levels in young worms, 1626 genes are higher in aged worms ($\log_2[\text{Fold-change (Day 1/Day 8)}] > 2.0$, p -adjusted < 0.001). (b) GO terms of genes that are expressed at higher levels in young (wild-type) whole animals highlight collagen and metabolism. (c) GO terms of genes that are expressed at higher levels in aged (wild-type) whole animals. GO terms were generated using Wormcat 2.0. (d) Tissue query for whole-worm aged-related genes highlights the alimentary system. (e) Full image of comparison of top wild-type neuronal and whole-worm differentially expressed with age genes. Related to **Figure 2D** [\[link\]](#).

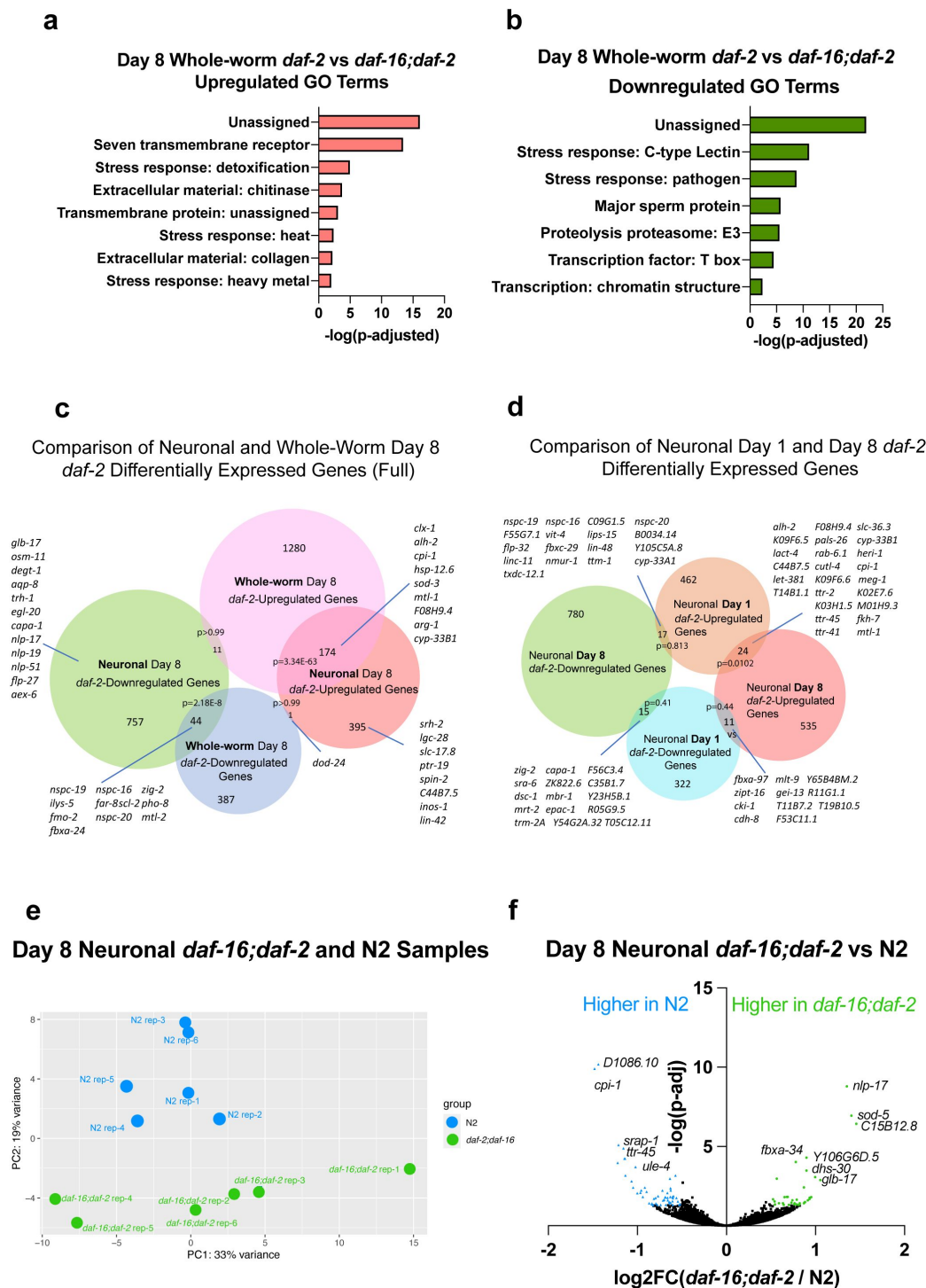


Figure S4.

Neuron-specific sequencing of Day 8 *daf-2* and *daf-16;daf-2* mutants.

(a-b) FACS results of neuron isolation. Over 99% of the cells collected are GFP+ neurons. 100,000 cells are collected for each biological replicate, 6 replicates for each genotype. (c) Representative image of N2 and *daf-2* lifespan. (d) Number graph of lifespan, learning, and memory function. Related to [Figure 4C](#) and [4D](#). (e) Number of genes detected in Day 8 N2, *daf-2*, and *daf-16;daf-2* neurons. (f) Ribosomal RNA depletion during sequencing. We used the library generation protocol with *C. elegans*-specific ribosomal RNA depletion kit (Tecan Genomics) successfully depleted rRNA to less than 20% of total reads. (g-h) Downsampling analysis for *daf-2* vs *daf-16;daf-2* and *daf-2* vs N2. Both shows sufficient depth.



Figure S5.

Whole-worm RNA-sequencing identifies changes in aged *daf-2* mutants.

(a-b) GO term analysis of whole-worm *daf-2*-regulated genes shows enrichment in stress-resistant genes. (c) Comparison of neuronal and whole-worm Day 8 *daf-2* differentially-expressed genes show high overlap (~30%), but also identify genes specific to neurons and to the whole body. Related to [Figure 3h](#). (d) Comparison of neuronal Day 1 and Day 8 *daf-2* differentially expressed genes identifies a set of consistent and sets of changed genes.

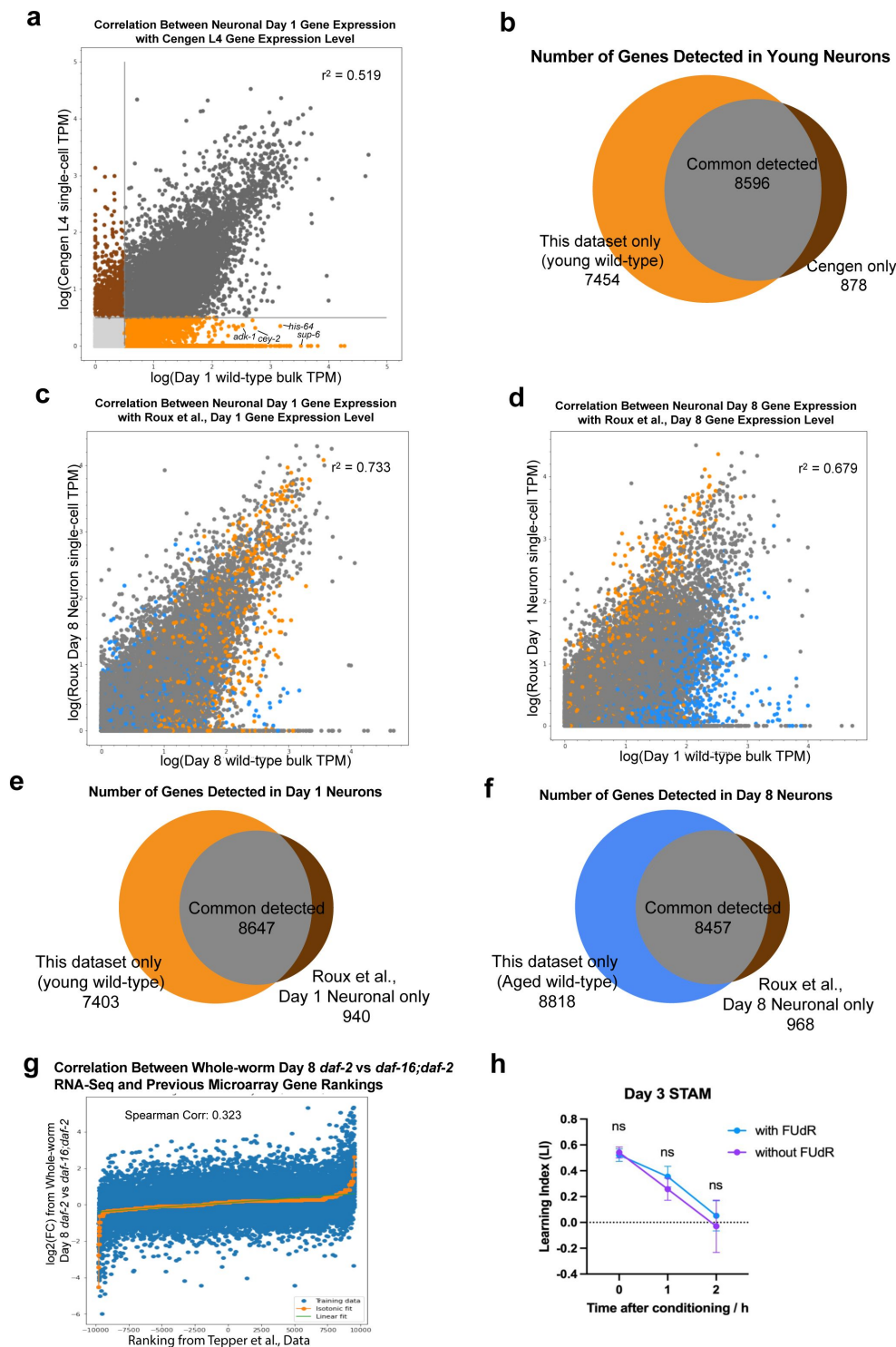


Figure S6.

DAF-16-dependent and -independent *daf-2*-regulated genes show different features.

(a) Volcano plot of *daf-2* vs N2 differentially expressed genes during aging. 1036 genes are more highly expressed in *daf-2* mutants, 1285 genes are higher in N2 ($\log_2[\text{Fold-change}(\textit{daf-2} \text{ vs N2})] > 0.5$, $p\text{-adjusted} < 0.05$). (b-c) Neuronal Day 8 *daf-2* vs N2 differentially expressed GO Terms. (d) Tissue query for *daf-2* vs *daf-16*; *daf-2* differentially expressed genes. (e) Tissue query for *daf-2* vs N2 differentially expressed genes. (f) Full image of comparison of neuronal Day 8 *daf-2* vs *daf-16*; *daf-2* and *daf-2* vs N2 differentially expressed genes. Related to **Figure 5B**.

References

1. Plassman B. L., et al. (2008) **Prevalence of cognitive impairment without dementia in the United States** *Ann Intern Med* **148**:427–434
2. Langa K. M., Levine D. A (2014) **The diagnosis and management of mild cognitive impairment: a clinical review** *JAMA* **312**:2551–2561
3. Gillis C., Mirzaei F., Potashman M., Ikram M. A., Maserejian N (2019) **The incidence of mild cognitive impairment: A systematic review and data synthesis.** *Alzheimer's & Dementia: Diagnosis Assessment & Disease Monitoring* **11**:248–256
4. Feigin V. L., et al. (2020) **The global burden of neurological disorders: translating evidence into policy** *The Lancet. Neurology* **19**:255–265
5. Nichols E., Vos T (2020) **Estimating the global mortality from Alzheimer's disease and other dementias: A new method and results from the Global Burden of Disease study 2019** *Alzheimer's & Dementia* **16**
6. White J., Southgate E., Thomson J., Brenner S (1986) **The structure of the nervous system of the nematode *Caenorhabditis elegans*.** *Philosophical transactions of the Royal Society of London. Series B Biological sciences* **314**:1–340
7. Kauffman A. L., Ashraf J. M., Corces-Zimmerman M. R., Landis J. N., Murphy C. T (2010) **Insulin Signaling and Dietary Restriction Differentially Influence the Decline of Learning and Memory with Age** *PLoS Biology* **8**
8. Hahm J. H., et al. (2015) ***C. elegans* maximum velocity correlates with healthspan and is maintained in worms with an insulin receptor mutation** *Nature Communications* **6**:1–7
9. Arey R. N., Murphy C. T (2017) **Conserved regulators of cognitive aging: From worms to humans** *Behavioural Brain Research* **322**:299–310
10. Kenyon C., Chang J., Gensch E., Rudner A., Tabtiang R (1993) **A *C. elegans* mutant that lives twice as long as wild type** *Nature* **366**:461–464
11. Murphy C. T., et al. (2003) **Genes that act downstream of DAF-16 to influence the lifespan of *Caenorhabditis elegans*** *Nature* **424**:277–284
12. Clancy D. J., et al. (2001) **Extension of life-span by loss of CHICO, a *Drosophila* insulin receptor substrate protein.** *Science (New York N.Y)* **292**:104–106
13. Blüher M., Kahn B. B., Kahn C. R (2003) **Extended longevity in mice lacking the insulin receptor in adipose tissue.** *Science (New York N.Y)* **299**:572–574
14. Suh Y., et al. (2008) **Functionally significant insulin-like growth factor I receptor mutations in centenarians** *Proceedings of the National Academy of Sciences of the United States of America* **105**:3438–3442
15. Willcox B. J., et al. (2008) **FOXO3A genotype is strongly associated with human longevity** *Proc Natl Acad Sci U S A* **105**:13987–13992

16. Liu J., et al. (2013) **Functional aging in the nervous system contributes to age-dependent motor activity decline in *C. elegans*** *Cell Metabolism* **18**:392–402
17. Li L. B., et al. (2016) **The Neuronal Kinesin UNC-104/KIF1A Is a Key Regulator of Synaptic Aging and Insulin Signaling-Regulated Memory** *Current biology : CB* **26**:605–615
18. Byrne A. B., et al. (2014) **Insulin/IGF1 Signaling Inhibits Age-Dependent Axon Regeneration** *Neuron* **81**:561–573
19. Lakhina V., et al. (2019) **ZIP-5/bZIP transcription factor regulation of folate metabolism is critical for aging axon regeneration** <https://doi.org/10.1101/727719>
20. Pan C. L., Peng C. Y., Chen C. H., McIntire S (2011) **Genetic analysis of age-dependent defects of the *Caenorhabditis elegans* touch receptor neurons** *Proceedings of the National Academy of Sciences of the United States of America* **108**:9274–9279
21. Tank E. M. H., Rodgers K. E., Kenyon C (2011) **Spontaneous Age-Related Neurite Branching in *Caenorhabditis elegans*** *J Neurosci* **31**:9279–9288
22. Toth M. L., et al. (2012) **Neurite Sprouting and Synapse Deterioration in the Aging *Caenorhabditis elegans* Nervous System** *J Neurosci* **32**:8778–8790
23. Bansal A., Zhu L. J., Yen K., Tissenbaum H. A (2015) **Uncoupling lifespan and healthspan in *caenorhabditis elegans* longevity mutants** *Proceedings of the National Academy of Sciences of the United States of America* **112**:E277–E286
24. Kaletsky R., et al. (2016) **The *C. elegans* adult neuronal IIS/FOXO transcriptome reveals adult phenotype regulators** *Nature* **529**:92–96
25. Gao S. M., et al. (2023) **Aging Atlas Reveals Cell-Type-Specific Regulation of Pro-longevity Strategies** *bioRxiv* <https://doi.org/10.1101/2023.02.28.530490>
26. Wang X., et al. (2022) **Ageing induces tissue-specific transcriptomic changes in *Caenorhabditis elegans*** *The EMBO Journal* **41**:1–20
27. Roux A. E., et al. (2023) **Individual cell types in *C. elegans* age differently and activate distinct cell-protective responses** *Cell Reports* **42**
28. Taylor S. R., et al. (2021) **Molecular topography of an entire nervous system** *Cell* **184**:4329–4347
29. Kauffman A., et al. (2011) ***C. elegans* positive butanone learning, short-term, and long-term associative memory assays** *Journal of Visualized Experiments* :1–9 <https://doi.org/10.3791/2490>
30. Stein G. M., Murphy C. T (2012) **The intersection of aging, longevity pathways, and learning and memory in *C. elegans*** *Frontiers in Genetics* **3**
31. Stein G. M., Murphy C. T (2014) ***C. elegans* positive olfactory associative memory is a molecularly conserved behavioral paradigm** *Neurobiology of Learning and Memory* **115**:86–94
32. Robinson D. G., Storey J. D (2014) **subSeq: Determining Appropriate Sequencing Depth Through Efficient Read Subsampling** *Bioinformatics* **30**:3424–3426

33. Artan M., et al. (2016) **Food-derived sensory cues modulate longevity via distinct neuroendocrine insulin-like peptides** *Genes Dev* **30**:1047–1057
34. Chen Z., et al. (2013) **Two insulin-like peptides antagonistically regulate aversive olfactory learning in *C. elegans*** *Neuron* **77**:572–585
35. Brear A. G., Yoon J., Wojtyniak M., Sengupta P (2014) **Diverse cell type-specific mechanisms localize G protein-coupled receptors to *Caenorhabditis elegans* sensory cilia** *Genetics* **197**:667–684
36. Savage C., et al. (1989) **mec-7 is a beta-tubulin gene required for the production of 15-protofilament microtubules in *Caenorhabditis elegans*** *Genes Dev* **3**:870–881
37. Maro G. S., et al. (2015) **MADD-4/Punctin and Neurexin Organize *C. elegans* GABAergic Postsynapses through Neuroligin** *Neuron* **86**:1420–1432
38. Vandamme J., et al. (2012) **The *C. elegans* H3K27 Demethylase UTX-1 Is Essential for Normal Development, Independent of Its Enzymatic Activity** *PLOS Genetics* **8**
39. Jin C., et al. (2011) **Histone demethylase UTX-1 regulates *C. elegans* life span by targeting the insulin/IGF-1 signaling pathway** *Cell Metab* **14**:161–172
40. Maures T. J., Greer E. L., Hauswirth A. G., Brunet A (2011) **The H3K27 demethylase UTX-1 regulates *C. elegans* lifespan in a germline-independent, insulin-dependent manner** *Aging Cell* **10**:980–990
41. Guillermo A. R. R., et al. (2021) **H3K27 modifiers regulate lifespan in *C. elegans* in a context-dependent manner** *BMC Biology* **19**
42. Shaw C. K., et al. (2023) **X Chromosome Factor Kdm6a Enhances Cognition Independent of Its Demethylase Function in the Aging XY Male Brain** *J Gerontol A Biol Sci Med Sci* **78**:938–943
43. Tang G.-B., et al. (2017) **The Histone H3K27 Demethylase UTX Regulates Synaptic Plasticity and Cognitive Behaviors in Mice** *Frontiers in Molecular Neuroscience* **10**
44. Fernández E. M., et al. (2022) **Neuronal membrane glycoprotein (nmgp-1) gene deficiency affects chemosensation-related behaviors, dauer exit and egg-laying in *Caenorhabditis elegans*** *J Neurochem* **160**:234–255
45. Tepper R. G., et al. (2013) **PQM-1 complements DAF-16 as a key transcriptional regulator of DAF-2-mediated development and longevity** *Cell* **154**:676–690
46. Gauthier S., Kaur G., Mi W., Tizon B., Levy E (2011) **Protective mechanisms by cystatin C in neurodegenerative diseases** *Front Biosci (Schol Ed)* **3**:541–554
47. Carmichael K., et al. (2021) **Function and Regulation of ALDH1A1-Positive Nigrostriatal Dopaminergic Neurons in Motor Control and Parkinson's Disease** *Front Neural Circuits* **15**
48. Li X., Masliah E., Reixach N., Buxbaum J. N (2011) **Neuronal Production of Transthyretin in Human and Murine Alzheimer's Disease: Is It Protective?** *J Neurosci* **31**:12483–12490
49. Lee H., et al. (2020) ***Xenopus* gpX3 Mediates Posterior Development by Regulating Cell Death during Embryogenesis** *Antioxidants (Basel)* **9**

50. Hambright W. S., Fonseca R. S., Chen L., Na R., Ran Q (2017) **Ablation of ferroptosis regulator glutathione peroxidase 4 in forebrain neurons promotes cognitive impairment and neurodegeneration** *Redox Biol* **12**:8–17
51. González-Calvo I., Cizeron M., Bessereau J.-L., Selimi F (2022) **Synapse Formation and Function Across Species: Ancient Roles for CCP, CUB, and TSP-1 Structural Domains** *Front Neurosci* **16**
52. Ung D. C., et al. (2018) **Ptchd1 deficiency induces excitatory synaptic and cognitive dysfunctions in mouse** *Mol Psychiatry* **23**:1356–1367
53. Perland E., et al. (2016) **The Putative SLC Transporters Mfsd5 and Mfsd11 Are Abundantly Expressed in the Mouse Brain and Have a Potential Role in Energy Homeostasis** *PLoS One* **11**
54. Huang Y., et al. (2023) **HSPB2 facilitates neural regeneration through autophagy for sensorimotor recovery after traumatic brain injury** *JCI Insight* **8**
55. Flynn J. M., Melov S (2013) **SOD2 in Mitochondrial Dysfunction and Neurodegeneration** *Free Radic Biol Med* **62** <https://doi.org/10.1016/j.freeradbiomed.2013.05.027>
56. Koukoulis F., Changeux J.-P (2020) **Do Nicotinic Receptors Modulate High-Order Cognitive Processing?** *Trends in Neurosciences* **43**:550–564
57. Zeiger J. S., et al. (2008) **The neuronal nicotinic receptor subunit genes (CHRNA6 and CHRNA3) are associated with subjective responses to tobacco** *Human Molecular Genetics* **17**:724–734
58. Lautrup S., Sinclair D. A., Mattson M. P., Fang E. F (2019) **NAD⁺ in Brain Aging and Neurodegenerative Disorders** *Cell Metabolism* **30**:630–655
59. Smies C. W., Bodinayake K. K., Kwapis J. L (2022) **Time to learn: The role of the molecular circadian clock in learning and memory** *Neurobiology of Learning and Memory* **193**
60. Lakhina V., et al. (2015) **Genome-wide functional analysis of CREB/long-term memory-dependent transcription reveals distinct basal and memory gene expression programs** *Neuron* **85**:330–345
61. Wang & Ruvkun, Gary (2004) **The insulin pathway regulates RNAi in C. elegans** *Cold Spring Harbor Symposium on Quantitative Biology* :429–433
62. Kaletsky R., et al. (2018) **Transcriptome analysis of adult Caenorhabditis elegans cells reveals tissue-specific gene and isoform expression** *PLoS Genetics* **14**
63. Shapira M., et al. (2006) **A conserved role for a GATA transcription factor in regulating epithelial innate immune responses** *Proceedings of the National Academy of Sciences* **103**:14086–14091
64. Eckl J., Sima S., Marcus K., Lindemann C., Richter K (2017) **Hsp90-downregulation influences the heat-shock response, innate immune response and onset of oocyte development in nematodes** *PloS one* **12**

65. Mack H. I. D., Kremer J., Albertini E., Mack E. K. M., Jansen-Dürr P (2022) **Regulation of fatty acid desaturase- and immunity gene-expression by mbk-1/DYRK1A in *Caenorhabditis elegans*** *BMC genomics* **23**
66. Murphy C. T., Lee S.-J., Kenyon C (2007) **Tissue entrainment by feedback regulation of insulin gene expression in the endoderm of *Caenorhabditis elegans*** *Proceedings of the National Academy of Sciences* **104**:19046–19050
67. Thomé-Duret V., et al. (1998) **Continuous glucose monitoring in the free-moving rat** *Metabolism* **47**:799–803
68. Kawano T., et al. (2000) **Molecular Cloning and Characterization of a New Insulin/IGF-like Peptide of the Nematode *Caenorhabditis elegans*** *Biochemical and Biophysical Research Communications* **273**:431–436
69. Pierce S. B., et al. (2001) **Regulation of DAF-2 receptor signaling by human insulin and ins-1, a member of the unusually large and diverse *C. elegans* insulin gene family** *Genes Dev* **15**:672–686
70. Li W., Kennedy S. G., Ruvkun G. (2003) **daf-28 encodes a *C. elegans* insulin superfamily member that is regulated by environmental cues and acts in the DAF-2 signaling pathway** *Genes Dev* **17**:844–858
71. Brenner S (1974) **THE GENETICS OF CAENORHABDITIS ELEGANS** *Genetics* **77**
72. Barrett A., et al. (2021) **A head-to-head comparison of ribodepletion and polyA selection approaches for *Caenorhabditis elegans* low input RNA-sequencing libraries** *G3 Genes | Genomes | Genetics* **11**
73. Love M. I., Huber W., Anders S (2014) **Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2** *Genome Biology* **15**
74. Raudvere U., et al. (2019) **g:Profiler: a web server for functional enrichment analysis and conversions of gene lists (2019 update)** *Nucleic Acids Research* **47**:W191–W198
75. Holdorf A. D., et al. (2020) **WormCat: An online tool for annotation and visualization of *caenorhabditis elegans* genome-scale data** *Genetics* **214**:279–294
76. Li Y., et al. (2018) **The Protective Role of Brain CYP2J in Parkinson’s Disease Models** *Oxidative Medicine and Cellular Longevity* **2018**
77. Ferguson C. S., Tyndale R. F (2011) **Cytochromes P450 in the brain: Emerging evidence for biological significance** *Trends Pharmacol Sci* **32**:708–714
78. Zhong L., et al. (2019) **Lipid transporter Spns2 promotes microglia pro-inflammatory activation in response to amyloid-beta peptide** *Glia* **67**:498–511
79. Cermak S., et al. (2016) **Loss of Cathepsin B and L Leads to Lysosomal Dysfunction, NPC-Like Cholesterol Sequestration and Accumulation of the Key Alzheimer’s Proteins** *PLOS ONE* **11**
80. Heremans I. P., et al. (2022) **Parkinson’s disease protein PARK7 prevents metabolite and protein damage caused by a glycolytic metabolite** *Proceedings of the National Academy of Sciences* **119**

81. Pastore S. F., Ko S. Y., Frankland P. W., Hamel P. A., Vincent J. B (2022) **PTCHD1: Identification and Neurodevelopmental Contributions of an Autism Spectrum Disorder and Intellectual Disability Susceptibility Gene** *Genes (Basel)* **13**
82. Lee K. M., Linskens A. M., Doe C. Q (2022) **Hunchback activates Bicoid in Pair1 neurons to regulate synapse number and locomotor circuit function** *Curr Biol* **32**:2430–2441
83. Broadie K., Baumgartner S., Prokop A (2011) **Extracellular matrix and its receptors in Drosophila neural development** *Dev Neurobiol* **71**:1102–1130
84. Fels E., et al. (2021) **Role of LGI1 protein in synaptic transmission: From physiology to pathology** *Neurobiol Dis* **160**
85. Rontani P., et al. (2021) **Impaired expression of the COSMOC/MOCOS gene unit in ASD patient stem cells** *Mol Psychiatry* **26**:1606–1618
86. Fu J., Guo O., Zhen Z., Zhen J (2020) **Essential Functions of the Transcription Factor Npas4 in Neural Circuit Development, Plasticity, and Diseases** *Frontiers in Neuroscience* **14**
87. **Neuronal PAS domain protein 4 (Npas4) controls neuronal homeostasis in pentylene-tetrazole-induced epilepsy through the induction of Homer1a** <https://doi.org/10.1111/jnc.14274>
88. Bernkopf M., et al. (2014) **Disruption of the methyltransferase-like 23 gene METTL23 causes mild autosomal recessive intellectual disability** *Hum Mol Genet* **23**:4015–4023
89. Reiff R. E., et al. (2014) **METTL23, a transcriptional partner of GABPA, is essential for human cognition** *Hum Mol Genet* **23**:3456–3466
90. Sinner D. I., Kim G. J., Henderson G. C., Igal R. A (2012) **StearoylCoA Desaturase-5: A Novel Regulator of Neuronal Cell Proliferation and Differentiation** *PLoS One* **7**
91. Hsu A., Podvin S., Hook V (2018) **Lysosomal Cathepsin Protease Gene Expression Profiles in Human Brain during Normal Development** *J Mol Neurosci* **65**:420–431
92. De Pasquale V., Moles A., Pavone L. M (2020) **Cathepsins in the Pathophysiology of Mucopolysaccharidoses: New Perspectives for Therapy** *Cells* **9**
93. Chuang Y.-C., Lee C.-H., Sun W.-H., Chen C.-C (2018) **Involvement of advillin in somatosensory neuron subtype-specific axon regeneration and neuropathic pain** *Proceedings of the National Academy of Sciences* **115**:E8557–E8566
1. Gauthier S., Kaur G., Mi W., Tizon B., Levy E (2011) **Protective mechanisms by cystatin C in neurodegenerative diseases** *Front Biosci (Schol Ed)* **3**:541–554
2. Carmichael K., et al. (2021) **Function and Regulation of ALDH1A1-Positive Nigrostriatal Dopaminergic Neurons in Motor Control and Parkinson's Disease** *Front Neural Circuits* **15**
3. Li X., Masliah E., Reixach N., Buxbaum J. N (2011) **Neuronal Production of Transthyretin in Human and Murine Alzheimer's Disease: Is It Protective?** *J Neurosci* **31**:12483–12490
4. Li Y., et al. (2018) **The Protective Role of Brain CYP2J in Parkinson's Disease Models** *Oxidative Medicine and Cellular Longevity* **2018**

5. Ferguson C. S., Tyndale R. F (2011) **Cytochromes P450 in the brain: Emerging evidence for biological significance** *Trends Pharmacol Sci* **32**:708–714
6. Zhong L., et al. (2019) **Lipid transporter Spns2 promotes microglia pro-inflammatory activation in response to amyloid-beta peptide** *Glia* **67**:498–511
7. Lee H., et al. (2020) **Xenopus gpx3 Mediates Posterior Development by Regulating Cell Death during Embryogenesis** *Antioxidants (Basel)* **9**
8. Hambright W. S., Fonseca R. S., Chen L., Na R., Ran Q (2017) **Ablation of ferroptosis regulator glutathione peroxidase 4 in forebrain neurons promotes cognitive impairment and neurodegeneration** *Redox Biol* **12**:8–17
9. Cermak S., et al. (2016) **Loss of Cathepsin B and L Leads to Lysosomal Dysfunction, NPC-Like Cholesterol Sequestration and Accumulation of the Key Alzheimer's Proteins** *PLOS ONE* **11**
10. Heremans I. P., et al. (2022) **Parkinson's disease protein PARK7 prevents metabolite and protein damage caused by a glycolytic metabolite** *Proceedings of the National Academy of Sciences* **119**
11. González-Calvo I., Cizeron M., Bessereau J.-L., Selimi F (2022) **Synapse Formation and Function Across Species: Ancient Roles for CCP, CUB, and TSP-1 Structural Domains** *Front Neurosci* **16**
12. Pastore S. F., Ko S. Y., Frankland P. W., Hamel P. A., Vincent J. B (2022) **PTCHD1: Identification and Neurodevelopmental Contributions of an Autism Spectrum Disorder and Intellectual Disability Susceptibility Gene** *Genes (Basel)* **13**
13. Ung D. C., et al. (2018) **Ptchd1 deficiency induces excitatory synaptic and cognitive dysfunctions in mouse** *Mol Psychiatry* **23**:1356–1367
14. Lee K. M., Linskens A. M., Doe C. Q (2022) **Hunchback activates Bicoid in Pair1 neurons to regulate synapse number and locomotor circuit function** *Curr Biol* **32**:2430–2441
15. Broadie K., Baumgartner S., Prokop A (2011) **Extracellular matrix and its receptors in Drosophila neural development** *Dev Neurobiol* **71**:1102–1130
16. Fels E., et al. (2021) **Role of LGI1 protein in synaptic transmission: From physiology to pathology** *Neurobiol Dis* **160**
17. Rontani P., et al. (2021) **Impaired expression of the COSMOC/MOCOS gene unit in ASD patient stem cells** *Mol Psychiatry* **26**:1606–1618
18. Perland E., et al. (2016) **The Putative SLC Transporters Mfsd5 and Mfsd11 Are Abundantly Expressed in the Mouse Brain and Have a Potential Role in Energy Homeostasis** *PLoS One* **11**
19. Fu J., Guo O., Zhen Z., Zhen J (2020) **Essential Functions of the Transcription Factor Npas4 in Neural Circuit Development, Plasticity, and Diseases** *Frontiers in Neuroscience* **14**
20. **Neuronal PAS domain protein 4 (Npas4) controls neuronal homeostasis in pentylenetetrazole-induced epilepsy through the induction of Homer1a** <https://doi.org/10.1111/jnc.14274>

21. Huang Y., et al. (2023) **HSPB2 facilitates neural regeneration through autophagy for sensorimotor recovery after traumatic brain injury** *JCI Insight* 8
22. Flynn J. M., Melov S (2013) **SOD2 in Mitochondrial Dysfunction and Neurodegeneration** *Free Radic Biol Med* 62 <https://doi.org/10.1016/j.freeradbiomed.2013.05.027>
23. Koukoulis F., Changeux J.-P (2020) **Do Nicotinic Receptors Modulate High-Order Cognitive Processing?** *Trends in Neurosciences* 43:550–564
24. Zeiger J. S., et al. (2008) **The neuronal nicotinic receptor subunit genes (CHRNA6 and CHRNA3) are associated with subjective responses to tobacco** *Human Molecular Genetics* 17:724–734
25. Bernkopf M., et al. (2014) **Disruption of the methyltransferase-like 23 gene METTL23 causes mild autosomal recessive intellectual disability** *Hum Mol Genet* 23:4015–4023
26. Reiff R. E., et al. (2014) **METTL23, a transcriptional partner of GABPA, is essential for human cognition** *Hum Mol Genet* 23:3456–3466
27. Sinner D. I., Kim G. J., Henderson G. C., Igal R. A (2012) **StearoylCoA Desaturase-5: A Novel Regulator of Neuronal Cell Proliferation and Differentiation** *PLoS One* 7
28. Lautrup S., Sinclair D. A., Mattson M. P., Fang E. F (2019) **NAD⁺ in Brain Aging and Neurodegenerative Disorders** *Cell Metabolism* 30:630–655
29. Smies C. W., Bodinayake K. K., Kwapis J. L (2022) **Time to learn: The role of the molecular circadian clock in learning and memory** *Neurobiology of Learning and Memory* 193
30. Hsu A., Podvin S., Hook V (2018) **Lysosomal Cathepsin Protease Gene Expression Profiles in Human Brain during Normal Development** *J Mol Neurosci* 65:420–431
31. De Pasquale V., Moles A., Pavone L. M (2020) **Cathepsins in the Pathophysiology of Mucopolysaccharidoses: New Perspectives for Therapy** *Cells* 9
32. Chuang Y.-C., Lee C.-H., Sun W.-H., Chen C.-C (2018) **Involvement of advillin in somatosensory neuron subtype-specific axon regeneration and neuropathic pain** *Proceedings of the National Academy of Sciences* 115:E8557–E8566

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Reviewer #1 (Public Review):

The authors perform RNA-seq on FACS isolated neurons from adult worms at days 1 and 8 of adulthood to profile the gene expression changes that occur with cognitive decline. Supporting data are included indicating that by day 7 of adulthood, learning and memory are reduced, indicating that this timepoint or after represents cognitively aged worms. Neuronal identity genes are reduced in expression within the cognitively aged worms, whereas genes involved in proteostasis, transcription/chromatin, and the stress response are elevated. A number of specific examples are provided, representing markers of specific neuronal subtypes, and correlating expression changes to the erosion of particular functions (e.g. motor neurons, chemosensory neurons, aversive learning neurons, etc).

To investigate whether upregulation of genes in neurons with age is compensatory or deleterious, the authors reduced expression of a set of three significantly upregulated genes and performed behavioral assays in young adults. In each case, reduction of expression improved memory, consistent with a model in which age-associated increases impair neuronal function.

The authors then characterize learning and memory in wild type, *daf-2*, and *daf-2/daf-16* worms with age and find that *daf-2* worms have an extended ability to learn for approximately 10 days longer than wild types. This was *daf-16* dependent. Memory was extended in *daf-2* as well, and strikingly, *daf-2;daf-16* had no short term memory even at day 1. Transcriptomic analysis of FACS-sorted neurons was performed on the three groups at day

8. The authors focus their analysis on *daf-2* vs. *daf-2;daf-16* and present evidence that *daf-2* neurons express a stress-resistance gene program. They also find small differences between the N2 and *daf-2;daf-16* neurons, which correlate with the observed behavioral differences, though these differences are modest.

The authors tested eight candidate genes that were more highly expressed in *daf-2* neurons vs. *daf-2;daf-16* and showed that reduction of 2 and 5 of these genes impaired learning and memory, respectively, in *daf-2* worms. This finding implicates specific neuronal transcriptional targets of IIS in maintaining cognitive ability in *daf-2* with age, which, importantly, are distinct from those in young wild type worms.

Overall, this is a strong study with rigorously performed experiments. The authors achieved their aim of identifying transcriptional changes in neurons that underlie loss of learning and memory in *C. elegans*, and how cognition is maintained in insulin/IGF-1-like signaling mutants.

<https://doi.org/10.7554/eLife.95621.2.sa2>

Reviewer #2 (Public Review):

Weng et al. perform a comprehensive study of gene expression changes in young and old animals, in wild-type and *daf-2* insulin receptor mutants, in the whole animal and specifically in the nervous system. Using this data, they identify gene families that are correlated with neuronal ageing, as well as a distinct set of genes that are upregulated in neurons of aged *daf-2* mutants. This is particularly interesting as *daf-2* mutants show both extended lifespan and healthier neurons in aged animals, reflected by better learning/memory in older animals compared with wild-type controls. Indeed, knockdown of several of these upregulated genes resulted in poorer learning and memory. In addition, the authors showed that several genes upregulated during ageing in wild-type neurons also contribute to learning and memory; specifically, knockdown of these genes in young animals resulted in improved memory. This indicates that (at least in this small number of cases), genes that show increased transcript levels with age in the nervous system somehow suppress memory, potentially by having damaging effects on neuronal health.

Finally, from a resource perspective, the neuronal transcriptome provided here will be very useful for *C. elegans* researchers as it adds to other existing datasets by providing the transcriptome of older animals (animals at day 8 of adulthood) and demonstrating the benefits of performing tissue-specific RNAseq instead of whole-animal sequencing.

The work presented here is of high quality and the authors present convincing evidence supporting their conclusions. I only have a few comments/suggestions:

(1) Do the genes identified to decrease learning/memory capacity in *daf-2* animals (Figure 4d/e) also impact neuronal health? *daf-2* mutant worms show delayed onset of age-related changes to neuron structure (Tank et al., 2011, J Neurosci). Does knockdown of the genes shown to affect learning also affect neuron structure during ageing, potentially one mechanism through which they modulate learning/memory?

(2) The learning and memory assay data presented in this study uses the butanone olfactory learning paradigm, which is well established by the same group. Have the authors tried other learning assays when testing for learning/memory changes after knockdown of candidate genes? Depending on the expression pattern of these genes, they may have more or less of an effect on olfactory learning versus for e.g. gustatory or mechanosensory-based learning.

(3) A comment on the 'compensatory vs dysregulatory' model as stated by the authors on page 7 - I understand that this model presents the two main options, but perhaps this is

slightly too simplistic: gene expression that rises during ageing may be detrimental for memory (= dysregulatory), but at the same time may also be beneficial other physiological roles in other tissues (=compensatory).

Comments on revised version:

I am satisfied with how the authors have addressed all my comments/suggestions.

<https://doi.org/10.7554/eLife.95621.2.sa1>

Reviewer #3 (Public Review):

Summary

In this manuscript, Weng et al. identify the neuron specific transcriptome that impacts age dependent cognitive decline. The authors design a pipeline to profile neurons from wild type and long-lived insulin receptor/IGF-1 mutants using timepoints when memory functions are declining. They discover signatures unique to neurons which validates their approach. The authors identify that genes related to neuronal identity are lost with age in wild type worms. For example, old neurons reduce the expression of genes linked to synaptic function and neuropeptide signaling and increase the expression of chromatin regulators, insulin peptides and glycoproteins. Depletion of selected genes which are upregulated in old neurons (*utx-1*, *ins-19* and *nmgp-1*) leads to improved short memory function. This indicates that some genes that increase with age have detrimental effects on learning and memory. The pipeline is then used to test neuronal profiles of long-lived insulin/IGF-1 *daf-2* mutants. Genes related to stress response pathways are upregulated in long lived *daf-2* mutants (e.g. *dod-24*, F08H9.4) and those genes are required for improved neuron function.

Strengths

The manuscript is well written, and the experiments are well described. The authors take great care to explain their reasoning for performing experiments in a specific way and guide the reader through the interpretation of the results, which makes this manuscript an enjoyable and interesting read. The authors discover novel regulators of learning and memory using neuron-specific transcriptomic analysis in aged animals, which underlines the importance of cell specific deep sequencing. The timepoints of the transcriptomic profiling are elegantly chosen, as they coincide with the loss of memory and can be used to specifically reveal gene expression profiles related to neuron function. The authors discuss on the *dod-24* example how powerful this approach is. In *daf-2* mutants whole-body *dod-24* expression differs from neuron specific profiles, which underlines the importance of precise cell specific approaches. This dataset will provide a very useful resource for the *C. elegans* and aging community as it complements existing datasets with additional time points and neuron specific deep profiling.

Weakness

This study nicely describes the neuron specific profiles of aged long-lived *daf-2* mutants. Selected neuronal genes that were upregulated in *daf-2* mutants (e.g. F08H9.4, *mtl-1*, *dod-24*, *alh-2*, C44B7.5) decreased learning/memory when knocked down. However, the knock down of these genes was not specific to neurons. The authors use a neuron-sensitive RNAi strain to address this concern and acknowledge this caveat in the text. While it is likely that selected candidates act only in neurons it is possible that other tissues participate as well.

<https://doi.org/10.7554/eLife.95621.2.sa0>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

*This important study investigates the transcriptional changes in neurons that underlie loss of learning and memory with age in *C. elegans*, and how cognition is maintained in insulin/IGF-1-like signaling mutants. The presented evidence is convincing, utilizing a cutting-edge method to isolate neurons from worms for genomics that is clearly conveyed with a rigorous experimental approach. Overall, this study supports that older *daf-2* worms maintain cognitive function via mechanisms that are unique from younger wild type worms, which will be of interest to neuroscientists and researchers studying ageing.*

Thank you, we appreciate the positive comments.

Public Reviews:**Reviewer #1 (Public Review):**

The authors perform RNA-seq on FACS-isolated neurons from adult worms at days 1 and 8 of adulthood to profile the gene expression changes that occur with cognitive decline. Supporting data are included indicating that by day 7 of adulthood, learning and memory are reduced, indicating that this time point or after represents cognitively aged worms. Neuronal identity genes are reduced in expression within cognitively aged worms, whereas genes involved in proteostasis, transcription/chromatin, and stress response are elevated. A number of specific examples are provided, representing markers of specific neuronal subtypes, and correlating expression changes to the erosion of particular functions (e.g. motor neurons, chemosensory neurons, aversive learning neurons, etc).

To investigate whether the upregulation of genes in neurons with age is compensatory or deleterious, the authors reduced the expression of a set of three significantly upregulated genes and performed behavioral assays in young adults. In each case, reduction of expression improved memory, consistent with a model in which age-associated increases impair neuronal function. This claim would be bolstered by an experiment elevating the expression of these genes in young neurons, which should reduce the learning index if the hypothesis is correct.

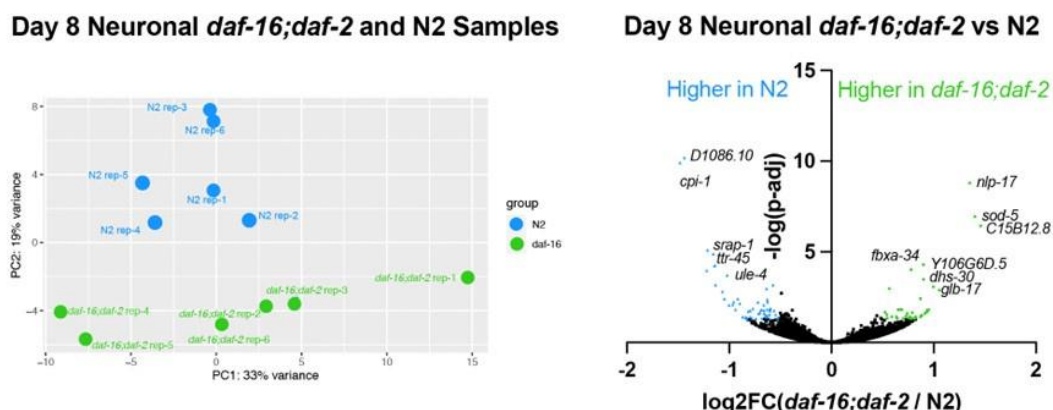
This is an interesting suggestion. Our long-term goal is to find ways to improve memory, and to better understand the “rules” that might govern changes with age. In this case, we were interested in addressing the hypothesis that genes that rise with age must be compensatory, which is a frequently stated theory that is not often tested. Here we showed that knocking down three genes that are upregulated in aged animals improved memory; our results suggest that the wild-type functions of these genes are likely deleterious for learning and memory functions, and further, that their increased expression with age is not a compensatory function. Certainly for future work, it might be interesting to better understand how and why these specific genes have a deleterious function that increases with age, and whether that function is different in younger animals where they are not highly expressed.

*The authors then characterize learning and memory in wild-type, *daf-2*, and *daf-2/daf-16* worms with age and find that *daf-2* worms have an extended ability to learn for approximately 10 days longer than wild types. This was *daf-16* dependent. Memory was*

extended in *daf-2* as well, and strikingly, *daf-2;daf-16* had no short-term memory even at day 1. Transcriptomic analysis of FACS-sorted neurons was performed on the three groups at day 8. The authors focus their analysis on *daf-2* vs. *daf-2;daf-16* and present evidence that *daf-2* neurons express a stress-resistance gene program. One question that remains unanswered is how well the N2 and *daf-2;daf-16* correlate overall, and are there differences? This may be informative as wild type and *daf-2;daf-16* mutants are not phenotypically identical when it comes to memory, and there may be differences that can be detected despite the overlap in the PCA. This analysis could reveal the *daf-16* targets involved in memory.

Re. *daf-2;daf-16* vs N2: This is a good suggestion. Our analysis in Fig. S5 showed that the *daf-2* vs N2 comparison shows similar results with the *daf-2* vs *daf-16;daf-2* comparison, but some additional genes are differentially expressed. Interestingly, the *daf-2* vs N2 comparison shows that the bZip transcription factors are upregulated in *daf-2* compared with N2 worms (Fig. S6f). This may indicate that additional transcription factors are controlled by the *daf-2* mutation in the nervous system in addition to the DAF-16/FOXO transcription factor.

Author response image 1.



We also identified the differentially expressed genes in the Day 8 neuronal *daf-16;daf-2* to N2 comparison, as the reviewer is asking about. The samples from different genotypes do separate from one another in the PCA plot, indicating there are differences between *daf-16;daf-2* and N2 neurons. However, the difference is smaller and there are fewer genes differentially expressed between *daf-16;daf-2* and N2: only 38 genes are significantly higher in *daf-16;daf-2*, and only 53 genes are significantly higher in N2 ($\log_2FC > 0.5$, $p\text{-adj} < 0.05$). The genes higher in N2 are enriched in endopeptidase inhibitors, and the genes higher in *daf-16;daf-2* are not enriched in any gene ontology terms. These results indicate that there are some differences between *daf-16;daf-2* and N2 neurons, which correlates with the behavioral differences we see, but the difference is small compared to *daf-2* neurons. We have added these data to the paper (Fig. S4e,f); thank you for the suggestion.

The authors tested eight candidate genes that were more highly expressed in *daf-2* neurons vs. *daf-2;daf-16* and showed that reduction of 2 and 5 of these genes impaired learning and memory, respectively, in *daf-2* worms. This finding implicates specific neuronal transcriptional targets of IIS in maintaining cognitive ability in *daf-2* with age, which, importantly, are distinct from those in young wild type worms.

Reviewer #2 (Public Review):

Weng et al. perform a comprehensive study of gene expression changes in young and old animals, in wild-type and daf-2 insulin receptor mutants, in the whole animal, and specifically in the nervous system. Using this data, they identify gene families that are correlated with neuronal ageing, as well as a distinct set of genes that are upregulated in neurons of aged daf-2 mutants. This is particularly interesting as daf-2 mutants show both extended lifespans and healthier neurons in aged animals, reflected by better learning/memory in older animals compared with wild-type controls. Indeed, the knockdown of several of these upregulated genes resulted in poorer learning and memory. In addition, the authors showed that several genes upregulated during ageing in wild-type neurons also contribute to learning and memory; specifically knockdown of these genes in young animals resulted in improved memory. This indicates that (at least in this small number of cases), genes that show increased transcript levels with age in the nervous system somehow suppress memory, potentially by having damaging effects on neuronal health.

Finally, from a resource perspective, the neuronal transcriptome provided here will be very useful for C. elegans researchers as it adds to other existing datasets by providing the transcriptome of older animals (animals at day 8 of adulthood) and demonstrating the benefits of performing tissue-specific RNAseq instead of whole-animal sequencing.

Thank you!

The work presented here is of high quality and the authors present convincing evidence supporting their conclusions.

Thanks!

I only have a few comments/suggestions:

(1) Do the genes identified to decrease learning/memory capacity in daf-2 animals (Figure 4d/e) also impact neuronal health? daf-2 mutant worms show delayed onset of age-related changes to neuron structure (Tank et al., 2011, J Neurosci). Does knockdown of the genes shown to affect learning also affect neuron structure during ageing, potentially one mechanism through which they modulate learning/memory?

Thank you for this suggestion, which would be good for a future direction, particularly for genes that might have some relationship to previously-identified cellular structural process. The genes we tested here include *dod-24*, *alh-2*, *mtl-1*, *F08H9.4*, *C44B7.5*, *hsp-12.3*, *hsp-12.6*, and *cpi-1*, which are related to stress response, proteolysis inhibitor, metabolic, and innate immunity GO categories, thus associated with stress resistance, proteolysis, lipid metabolism processes; none are obvious choices for morphological effects.

However, it is worth noting that learning and memory decline much faster (Days 4-8) than morphological differences are observed (generally after Day 12-15). Moreover, those morphological differences have been studied primarily in mechanosensory neurons (touch neurons) rather than the chemosensory neurons that are involved in learning and memory, so additional genes may be required for those differences that we were not focusing on in this study.

(2) The learning and memory assay data presented in this study uses the butanone olfactory learning paradigm, which is well established by the same group. Have the authors tried other learning assays when testing for learning/memory changes after the knockdown of candidate genes? Depending on the expression pattern of these genes,

they may have more or less of an effect on olfactory learning versus for example gustatory or mechanosensory-based learning.

The reason that we use the butanone olfactory learning paradigm is because it is more similar to learning of information (neutral odorant association with positive cue (food)) – the kind of memory we would like to preserve in humans - rather than a stress-induced memory, such as starvation or pathogenesis-associated aversive learning paradigms, which are more like PTSD. (There is likely to be quite a bit of overlap in mechanism, however, including the role of genes such as *magi-1* and *casy-1*, so it would not be surprising if many of these genes also were required for other learning paradigms.)

(3) I have a comment on the 'compensatory vs dysregulatory' model as stated by the authors on page 7. I understand that this model presents the two main options, but perhaps this is slightly too simplistic: the gene expression that rises during ageing may be detrimental for memory (= dysregulatory), but at the same time may also be beneficial for other physiological roles in other tissues (=compensatory).

This is a good point, and we made the clarification that in the text: “There may be other scenarios in which a gene with multiple functions may be detrimental for some behaviors but beneficial for other physiological roles.”

Reviewer #3 (Public Review):

Summary:

*In this manuscript, Weng et al. detect a neuron-specific transcriptome that regulates aging. The authors first profile neuron-specific responses during aging at a time point where a loss in memory function is present. They discover signatures unique to neurons which validate their pipeline and reveal the loss of neuron identity with age. For example, old neurons reduce the expression of genes related to synaptic function and neuropeptide signaling and increase the expression of chromatin regulators, insulin peptides, and glycoproteins. The authors discover the detrimental effect of selected upregulated genes (*utx-1*, *ins-19*, and *nmgp-1*) by knocking them down in the whole body and detecting improvement of short memory functions. They then use their pipeline to test neuronal profiles of long-lived insulin/IGF mutants. They discover that genes related to stress response pathways are upregulated upon longevity (e.g. *dod-24*, *F08H9.4*) and that they are required for improved neuron function in long-lived individuals.*

Strengths:

*Overall, the manuscript is well-written, and the experiments are well-described. The authors take great care to explain their reasoning for performing experiments in a specific way and guide the reader through the interpretation of the results, which makes this manuscript an enjoyable and interesting read. Using neuron-specific transcriptomic analysis in aged animals the authors discover novel regulators of learning and memory, which underlines the importance of cell-specific deep sequencing. The time points of the transcriptomic profiling are elegantly chosen, as they coincide with the loss of memory and can be used to specifically reveal gene expression profiles related to neuron function. The authors showcase on the *dod-24* example how powerful this approach is. In long-lived insulin/IGF-1 receptor mutants body-wide *dod-24* expression differs from neuron-specific profiles. Importantly, the depletion of *dod-24* has an opposing effect on lifespan and learning memory. The dataset will provide a useful resource for the *C. elegans* and aging community.*

Thank you, we do hope people will find the data useful.

Weaknesses:

While this study nicely describes the neuron-specific profiles, the authors do not test the relevance in a tissue-specific way. It remains unclear if modifying the responses only in neurons has implications for either memory or potentially for lifespan. The authors point to this in the text and refer to tissue-specific datasets. However, it is possible that the tissue-specific profile changes with age. The authors should consider mining publicly available cell-specific aging datasets and performing neuron-specific RNAi to test the functional relevance of the neuron-specific response. This would strengthen the importance of cell-specific profiling.

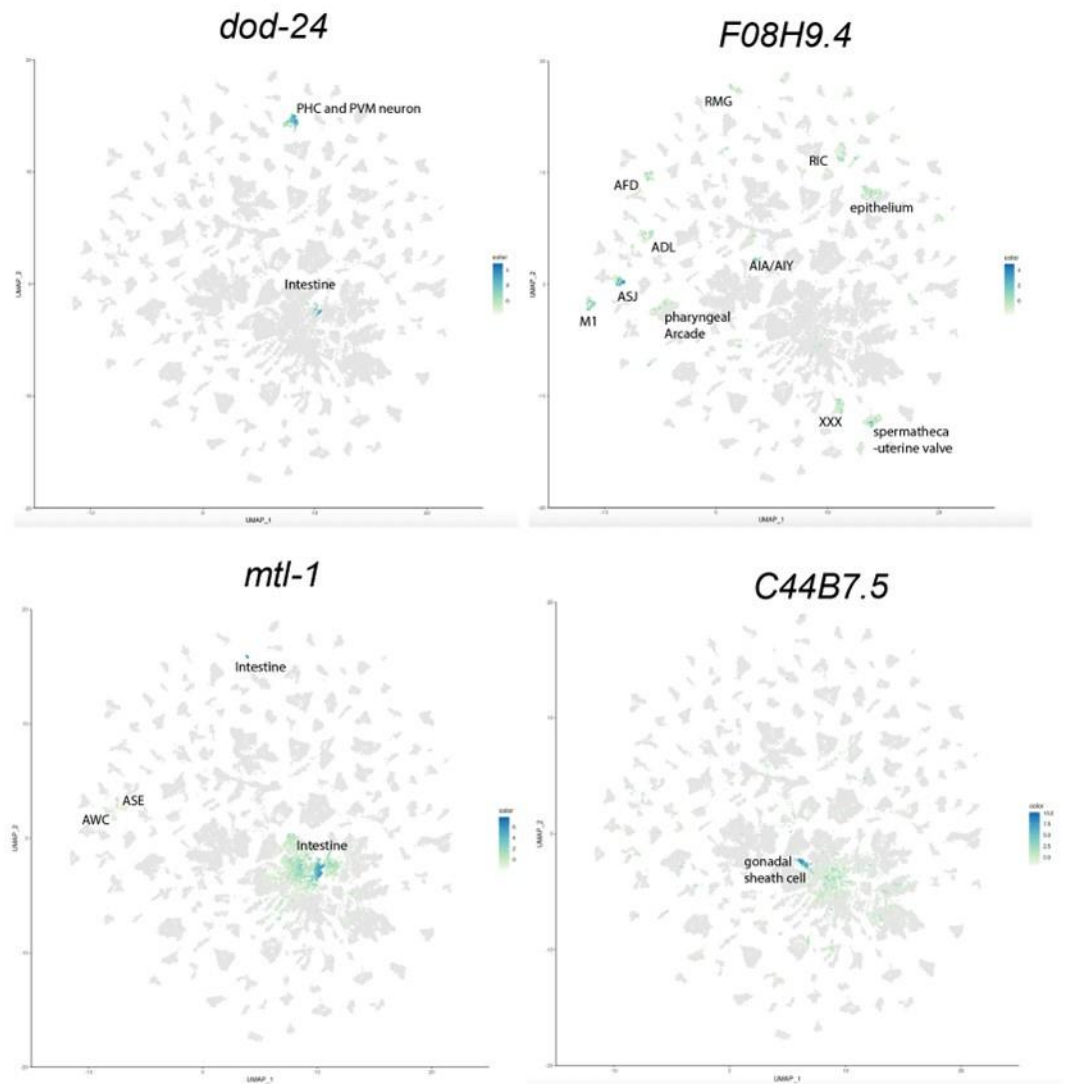
Thank you for your suggestions. As we have mentioned in the text, our candidate genes are either (1) only expressed in the neurons (*alh-2* and *F08H9.4*), or they are only more highly expressed in *daf-2* compared to wild type only in the nervous system (*C44B7.5* or *dod-24*). Thus, the effect we see from knocking down these genes in *daf-2* are likely neuron-specific. Additionally, we performed our assays with neuron-sensitive RNAi strain CQ745: *daf-2(e1370) III; vIs69 [pCFJ90(Pmyo-2::mCherry + Punc-119::sid-1)] V*. It has been previously shown that neuronal expression of *sid-1* decreases non-neuronal RNAi, suggesting that neurons expressing transgenic *sid-1(+)* served as a sink for dsRNA (Calixto et al., 2010). Thus, this neuron-sensitive RNAi is likely neuron-specific and our results is unlikely from knocking down these genes in non-neuronal tissues. However, we do acknowledge this issue.

To identify the expression pattern of these genes in a more cell-specific way in the adults, we examined the expression of our candidate genes that affected learning and memory, namely *dod-24*, *F08H9.4*, *C44B7.9*, *alh-2*, and *mtl-1*, in the Calico database (Roux et al., 2023). From that database, we can see that *dod-24* is mainly expressed in the PHC and PVM neurons, and *F08H9.4* is largely expressed in various neurons. Both have only slight expression outside the nervous system. *C44B7.5* and *mtl-1* are more broadly expressed, but *C44B7.5* was not found to be differentially expressed in other tissues in *daf-2*, and *mtl-1* only had a slight effect on learning and memory. Perhaps due to their sequencing depth and detection limit, Roux et al. didn't detect *alh-2* expression anywhere in their data.

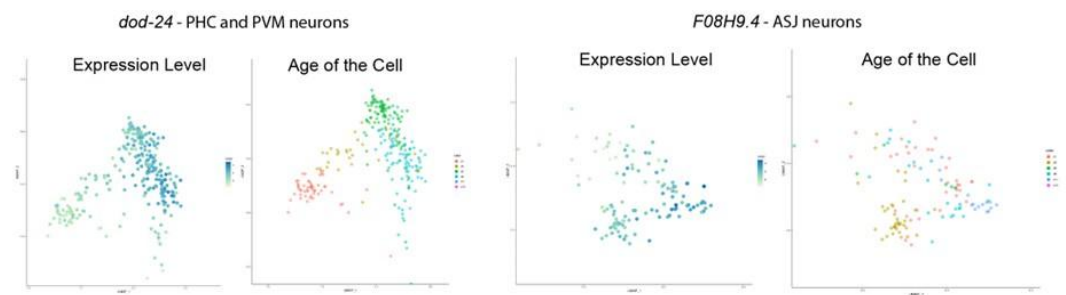
Thus, the neuron-specific expression and *daf-2* differential expression pattern of these genes indicate that the learning and memory improvement in aged *daf-2* is unlikely due to neuronal non-autonomous effects.

To better address this concern (that for the genes that we found only expressed in the neurons, the neuron-confined expression may change with age) we examined the expression pattern change of these genes with age. As is shown below, from the Calico database, we can see that the expression in the nervous system persists, and even slightly increases, with age, thus age-related expression pattern change is not a concern to our analysis.

Author response image 2.



Author response image 3.



Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Most of my comments are in the public section. A few additional recommendations for the authors regarding the formatting/presentation:

The presentation of Figure S6e-h in the introduction is somewhat confusing and feels out of order. If presented first, it should be S1. Otherwise, discussion of this figure should go at the end of the results section or in the discussion if appropriate.

Thank you for pointing this out. We have moved the discussion of this figure to the Discussion section.

I do not see Figure S5 described in the text.

Good catch, thank you. We have added the descriptions for Figure S5 in the text.

In general, check the figures, figure legends, and how they are referenced in the text, particularly the supplemental figures and legends.

Minor comments:

There is a typo in the Figure 4 legend: Neuronal IIX should be IIS.

Thanks for pointing this out. We have corrected it in the text.

Reviewer #2 (Recommendations For The Authors):

• There are multiple instances throughout the manuscript where there are statements in brackets that provide justification or explanation for some of the approaches used. There is no reason for 'side note' brackets to be used. I suggest removing them and incorporating these statements into the narrative.

Thank you, we have now incorporated these points into the main text.

• Introduction: page 4 "here we RNA-sequenced FACS-isolated neurons" should be "here we performed RNA sequencing on FACS-isolated neurons..."

Thank you, we have changed the text accordingly.

• Figure 2A: I do not understand the legend for this panel "Tissue Query for wild-type genes expressed at higher levels in aged worms show lower nervous system and neuron prediction score." Please clarify.

We have clarified the Figure 2A legend:

(A) Tissue prediction score for wild-type genes expressed at higher levels in aged worms.

• Page 8: "We previously observed that loss of single genes that play a role in complex behaviors like learning and memory can have a large impact on function 60, unlike the additive roles of longevity-promoting genes 11." - a large impact on what function?

Thank you for noting, we have clarified it in the text accordingly:

“We previously observed that for genes that play a role in complex behaviors like learning and memory, the loss of single genes can have a large impact on these complex behaviors 60, unlike the additive roles of longevity-promoting genes 11.”

• Next line “Therefore, one mechanism by which wild-type worms lose their function with age...” - again, what function?

Thank you for noting this, we have clarified the text to say we refer to the learning and memory functions.

• Page 9: “Thus, *daf-2* mutants maintain their higher cognitive quality of life longer than wild-type worms, while *daf-16*;*daf-2* mutants spend their whole lives without memory ability (Figure 3d), in contrast to claims that *daf-2* mutants are less healthy than wild-type or *daf-16* worms23.” - since ref 23 did not perform any learning/memory tests, the definition of ‘health’ in ref 23 is different to ‘cognitive health’ as studied here. So the findings in this study are not ‘in contrast’ to ref 23 but rather add to these findings.

Learning and memory ability is an important function for a healthy individual, thus we would assert that indeed, cognitive health is an important part of the “health” of *daf-2* worms. In ref 23, Bansal et al. claim that *daf-2* worms are less healthy without assessing their learning and memory ability; their lack of data is an insufficient reason for us to remove our statement, as cognitive health is part of healthspan. Here we find that the “learning span” of *daf-2* lasts at least proportionally if not longer than that of wild type. We have also previously shown that *daf-2* worms also have longer maximum velocity span with age (Hahm et al., 2015), in direct contrast with Bansal et al.’s claim that *daf-2* worms move less well and thus are less healthy – *daf-2* worms simply stop sooner when presented with food and switch to feeding, due to their higher *odr-10* levels. The Bansal paper continues to be frequently cited as finding that *daf-2* mutants are less healthy than wild type, a claim for which we can still find no experimental evidence to support. Therefore, it is important that we make the point that *daf-2* worms have extended cognitive health, which is part of health span.

• Page 13: I feel like the sentence “Furthermore, memory maintenance with age might require additional functions that were not previously uncovered in analyses of young animals” is both vague (what functions are referred to?) and a little bit obvious (obvious that age-related changes would not be revealed in analyses of young animals). Perhaps rephrase to make the desired point clearer?

We have clarified the sentence in the text:

“Furthermore, memory maintenance with age might require additional genes that function in promoting stress resistance and neuronal resilience, which were not previously uncovered in analyses of young animals.”

<https://doi.org/10.7554/eLife.95621.2.sa4>